Journal of Industrial Intelligence
https://www.acadlore.com/journals/JII

Cross-Condition Remaining Useful Life Prediction of Rolling Bearings via Multi-Source Weighted Transfer Learning and Subdomain Alignment

Yilei Cui*, Luqi Zhang

College of Mechanical and Electrical Engineering, Nanjing University of Aeronautics and Astronautics, 210016 Nanjing, China

* Correspondence: Yilei Cui (cuiyilei0123@163.com)

Received: 02-22-2026 **Revised:** 04-09-2026 **Accepted:** 04-24-2026

Citation: Y. L. Cui and L. Q. Zhang, "Cross-condition remaining useful life prediction of rolling bearings via multi-source weighted transfer learning and subdomain alignment," *J. Ind Intell.*, vol. 4, no. 2, pp. 93–130, 2026. https://doi.org/10.56578/jii040202.

© 2026 by the author(s). Published by Acadlore Publishing Services Limited, Hong Kong. This article is available for free download and can be reused and cited, provided that the original published version is credited, under the CC BY 4.0 license.

Abstract: Accurate remaining useful life (RUL) prediction of rolling bearings is essential for predictive maintenance and reliable operation of industrial rotating machinery. However, variations in operating conditions introduce substantial differences in vibration characteristics and degradation patterns, limiting the applicability of models trained under individual operating conditions. This study investigates a multi-source transfer learning approach for bearing RUL prediction across different operating conditions. A framework termed deep residual shrinkage network with multi-source weighted screening (DRSN-MW) was developed by integrating a deep residual shrinkage network with multi-source domain screening, subdomain alignment, and physical monotonicity constraints. An adaptive feature extraction module was constructed to capture degradation characteristics from vibration signals, while weighted extension correlation and dynamic time warping (DTW) were combined to identify source domains with similar degradation patterns. Degradation-stage partitioning and entropy-weighted local maximum mean discrepancy (LMMD) were employed to align source and target features within corresponding degradation stages. Physical monotonicity and decision consistency constraints were further incorporated to regulate the predicted RUL trajectories. The framework was evaluated using the 2012 Prognostics and Health Management (PHM2012) bearing dataset and Xi'an Jiaotong University–Sumyoung (XJTU-SY) bearing dataset under single-source and multi-source transfer scenarios. Average mean absolute errors (MAEs) of 0.126 and 0.129 and root mean square errors (RMSEs) of 0.150 and 0.156 were obtained on the two datasets, respectively. On the PHM2012 dataset, the proposed method reduced the average root mean square error by 33.6% compared with the strongest baseline in the reported comparison. The results indicate that combining source-domain screening with degradation-stage alignment and physical constraints can improve RUL prediction across operating conditions. The proposed framework provides a data-driven approach to bearing health prognostics and predictive maintenance in industrial rotating machinery.

Keywords: Remaining useful life prediction; Rolling bearings; Predictive maintenance; Multi-source transfer learning; Subdomain alignment; Deep residual shrinkage network

1 Introduction

Rolling element bearings are essential components of industrial rotating machinery and are widely used in manufacturing equipment, power generation systems, and other mechanical installations [1]. During prolonged operation, bearings are subjected to varying rotational speeds, mechanical loads, and thermal conditions, which may accelerate material degradation and eventually lead to mechanical failure. As components particularly susceptible to wear and fatigue, their operating conditions directly influence equipment reliability, operational safety, and maintenance costs [2]. Bearing-related failures account for a substantial proportion of unexpected machinery downtime in industrial applications [3]. Consequently, accurate prediction of bearing remaining useful life (RUL) is an important element of condition-based maintenance (CBM), allowing maintenance activities to be planned according to equipment degradation rather than predetermined service intervals [4].

Despite considerable progress in machinery health prognostics, reliable RUL prediction under varying operating conditions remains difficult. Industrial rotating machinery frequently operates under changing rotational speeds,

mechanical loads, and temperatures, resulting in variations in vibration responses and degradation characteristics [5]. These changes may alter both the statistical distributions of monitoring signals and the progression of bearing deterioration. Moreover, obtaining complete run-to-failure data for every operating condition is generally impractical because accelerated life testing requires considerable time and experimental resources [6, 7]. Consequently, the available degradation records often cover only a limited range of operating conditions, while data collected from different machines or operating environments may exhibit substantial distribution discrepancies. Models developed using data from a single operating condition may therefore produce unreliable predictions when applied to previously unseen conditions [8]. Addressing these discrepancies is essential for developing RUL prediction methods suitable for industrial predictive maintenance.

Transfer learning provides a means of addressing this problem by transferring degradation knowledge from operating conditions with sufficient labeled data to target conditions for which labeled observations are limited or unavailable [9, 10]. Within this field, deep domain adaptation (DDA) has received considerable attention because it allows transferable representations to be learned directly from monitoring signals. Existing DDA approaches can broadly be divided into feature distribution alignment methods and adversarial adaptation methods, depending on how discrepancies between source and target domains are addressed.

Feature alignment methods generally seek to reduce statistical differences between source and target representations while retaining information relevant to the prediction task. Maximum mean discrepancy (MMD) has been widely adopted for this purpose, and studies by Zhang et al. [11] and Wang et al. [12] have provided methodological foundations for deep feature learning and machinery prognostics. Zhu et al. [13] further investigated transfer learning for bearing monitoring under different operating conditions. Nevertheless, alignment based primarily on global feature distributions may not adequately account for differences among successive stages of bearing degradation. Early degradation is often characterized by relatively weak changes in vibration responses, whereas advanced deterioration may involve more pronounced fault-related impulses and rapidly changing signal characteristics. These stages may exhibit different degradation mechanisms and feature distributions [14]. Consequently, global alignment alone may produce inappropriate correspondence between source and target samples at different degradation stages, particularly when their degradation trajectories progress at different rates [15].

Adversarial domain adaptation offers another approach by learning feature representations that are less sensitive to differences between operating conditions [16–18]. Related studies have investigated adversarial training, attention mechanisms, and feature disentanglement for machinery fault diagnosis and health monitoring. For example, Li et al. [19] examined deep-learning-based representations for rotating machinery, while Hu et al. [20] investigated feature disentanglement for bearing RUL prediction under different working conditions. Although these approaches provide useful strategies for cross-domain feature learning, adversarial optimization may introduce additional training complexity and sensitivity to model configuration. A further difficulty arises when several source domains are available. Multi-source transfer learning methods [21–23] can incorporate information from multiple operating conditions, but differences in degradation patterns mean that not every source domain contributes equally to the target prediction task. Source selection based only on general distribution similarity may overlook differences in degradation progression, increasing the possibility of negative transfer when poorly matched source domains are included in model training.

Recent research has also examined self-supervised learning and pretrained representations as alternatives for reducing dependence on labeled degradation data. Liu et al. [24] investigated learning strategies for bearing fault prediction, while Zhou and Barati Farimani [25] developed a Transformer-based pretraining approach for transferable bearing fault representations. Such developments indicate the potential of representation learning for extracting useful information from vibration measurements. However, when target-domain pseudo-labels are used to guide adaptation, errors introduced during the initial prediction stage may propagate through subsequent training iterations [26]. Studies by Chen et al. [27] and Peng et al. [28] have also explored feature alignment and domain adaptation in related learning problems, providing further methodological references for cross-domain representation learning. Despite these developments, the integration of degradation characteristics and physically meaningful constraints into transfer learning remains an important issue in machinery prognostics [29]. In particular, models optimized primarily according to prediction error may produce fluctuations or locally increasing RUL estimates that are inconsistent with the expected progression of bearing degradation. Such behavior can complicate the interpretation of predicted degradation trajectories and their use in industrial maintenance planning.

The preceding studies indicate three issues that require further attention in cross-condition bearing RUL prediction. First, global domain adaptation may overlook differences between early and advanced degradation stages, resulting in incomplete alignment of degradation-related features. Second, the selection and combination of multiple source domains require greater consideration of degradation trend similarity to limit the influence of unrelated operating conditions. Third, prediction models need to account for the temporal consistency of bearing degradation so that estimated RUL trajectories remain physically reasonable. These issues are closely related: unsuitable source-domain selection can introduce inconsistent degradation information, while insufficient stage-wise

alignment and unconstrained regression may further increase prediction errors under changing operating conditions.

To address these issues, this study developed a deep residual shrinkage network with multi-source weighted screening (DRSN-MW) for cross-condition RUL prediction of rolling bearings. The framework combined noise-robust feature extraction, weighted source-domain screening, degradation-stage alignment, and physically constrained prediction within a unified transfer learning architecture. Its main contributions are summarized as follows.

1. A noise-robust degradation feature extraction module was developed using a DRSN with channel-adaptive threshold adjustment and multi-scale feature fusion. The module was designed to suppress noise-related responses while retaining vibration features associated with different stages of bearing degradation under varying operating conditions.
2. A multi-source domain screening strategy was established by combining entropy-weighted extension correlation with dynamic time warping (DTW). The strategy evaluated both feature similarity and temporal degradation characteristics to identify source domains more closely related to the target operating condition, thereby reducing the influence of potentially unsuitable source data during knowledge transfer.
3. A degradation-stage-aware subdomain alignment method was developed using adaptive stage partitioning, Mix-up-based feature recombination, and entropy-weighted local MMD. The method aligned source and target representations within corresponding degradation stages rather than relying exclusively on global distribution matching, allowing differences in stage-dependent degradation characteristics to be considered during model adaptation.
4. Decision consistency regularization and a physical monotonicity constraint were incorporated into the prediction objective to regulate the outputs of multiple source-domain regression heads. These constraints were intended to reduce inconsistencies among regression predictions and discourage increases in estimated RUL along the degradation trajectory, supporting more stable and physically consistent bearing life predictions.

The proposed framework was evaluated using the 2012 Prognostics and Health Management (PHM2012) bearing dataset and Xi'an Jiaotong University–Sumyoung (XJTU-SY) bearing dataset under single-source and multi-source cross-condition transfer scenarios. Prediction accuracy, the contribution of individual model components, and resistance to signal noise were examined through comparative experiments and ablation studies. The overall research procedure is presented in Figure 1.

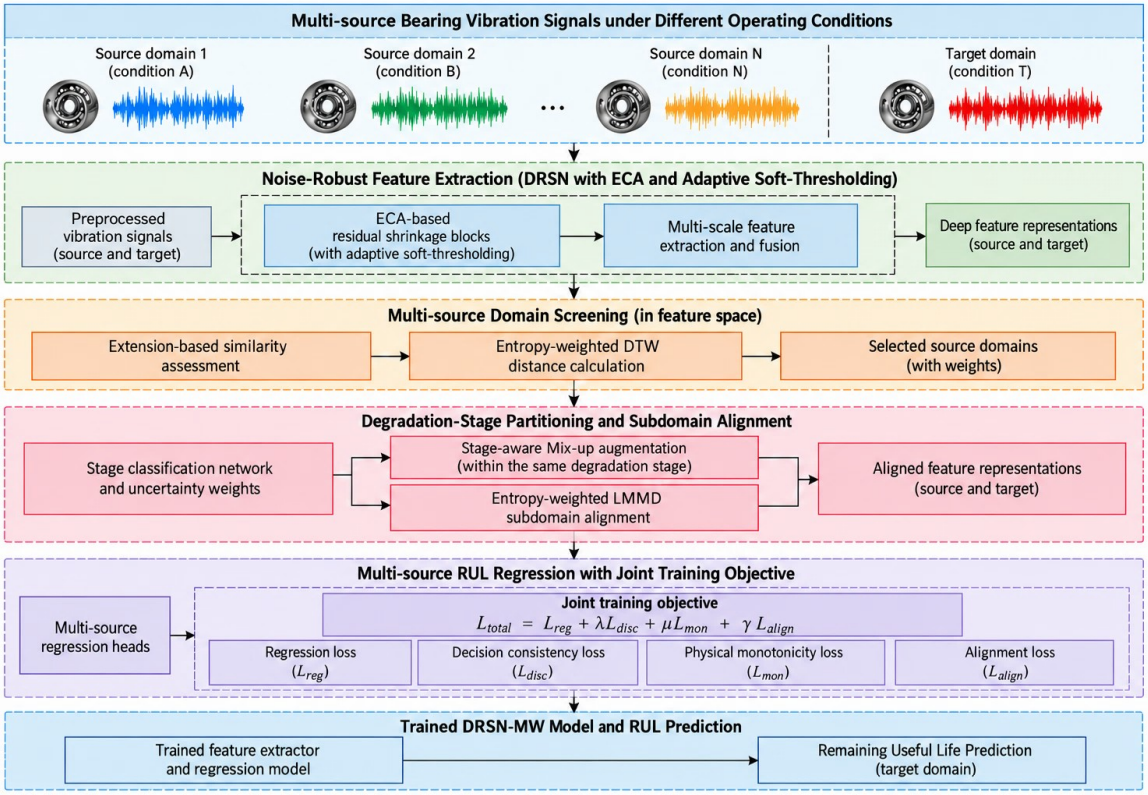


Figure 1. Overview of the proposed DRSN-MW framework for cross-condition bearing RUL prediction

Note: DRSN-MW = deep residual shrinkage network with multi-source weighted screening; RUL = remaining useful life; DRSN = deep residual shrinkage network; ECA = efficient channel attention; DTW = dynamic time warping; LMMD = local maximum mean discrepancy.

2 Preliminaries

2.1 Problem Description

Cross-condition RUL prediction involves transferring degradation knowledge from one or more operating conditions to a target condition with a different data distribution. Domain adaptation generally assumes that the marginal distributions of the source and target domains differ,

$$P(X_s) \neq P(X_t)$$

while their conditional distributions remain sufficiently similar,

$$P(Y_s \mid X_s) \approx P(Y_t \mid X_t)$$

to permit knowledge transfer. In bearing prognostics, however, changes in rotational speed, mechanical load, and degradation progression may affect both the statistical characteristics of vibration signals and their relationships with the corresponding RUL values. Consequently, alignment of global feature distributions alone may not adequately preserve the degradation information required for accurate prediction.

As illustrated in Figure 2a, global domain adaptation attempts to reduce the overall distribution discrepancy between source and target features without explicitly distinguishing individual degradation stages. This may cause samples representing different health conditions to become incorrectly aligned in the latent feature space, particularly when the source and target bearings exhibit different degradation rates. Such misalignment can affect the learned regression function and introduce errors into the predicted RUL trajectories.

Subdomain adaptation addresses this limitation by considering the internal structure of the degradation process during feature alignment. Rather than treating all observations as belonging to a single distribution, the bearing degradation process is divided into corresponding health stages, and feature distributions are aligned within these stages. As shown in Figure 2b, this approach seeks to reduce inter-domain discrepancies while retaining the distinctions between successive degradation states. Stage-wise alignment is therefore particularly relevant to cross-condition bearing prognostics, where the characteristics of early degradation and advanced fault development may differ substantially.

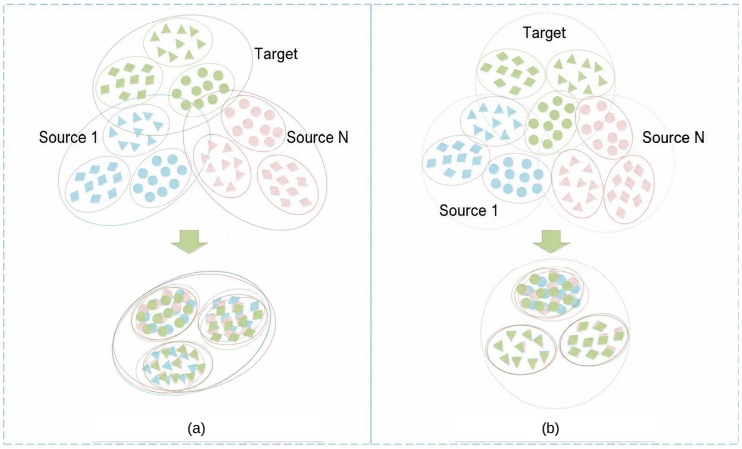


Figure 2. Comparison of domain adaptation strategies: (a) global distribution alignment; (b) degradation-stage-based subdomain alignment

For the multi-source cross-condition RUL prediction problem considered in this study, labeled bearing degradation data collected under N operating conditions were used to construct the source-domain set, denoted by

$$\mathcal{D}_s = \left\{\mathcal{D}_s^1, \mathcal{D}_s^2, \ldots, \mathcal{D}_s^N\right\}$$

where, N is the number of available source domains. The target domain is denoted by $\mathcal{D}_t$.

The i-th source domain is defined as

$$\mathcal{D}_s^i = \left\{\left(x_{s,j}^i, y_{s,j}^i\right)\right\}_{j=1}^{n_s^i}$$

where, $x_{s,j}^i$ represents the vibration sample indexed by j in source domain i, $y_{s,j}^i$ is its corresponding RUL label, and n_s^i denotes the number of samples in that domain. The source-domain observations follow a joint probability distribution P_s^i.

Similarly, the target domain is expressed as

$$\mathcal{D}_t = \{x_{t,j}\}_{j=1}^{n_t}$$

where, $x_{t,j}$ is the j-th target-domain vibration sample and n_t is the number of target-domain samples. The target observations follow a distribution P_t, and their RUL labels are unavailable during model training. The objective is to learn transferable degradation representations from the labeled source domains and unlabeled target-domain observations, allowing RUL predictions to be generated for bearings operating under the target condition.

2.2 Deep Residual Shrinkage Networks

The DRSN [30] is a residual learning architecture designed to extract informative features from signals affected by noise and interference. It incorporates adaptive soft-thresholding into residual blocks, allowing low-magnitude feature responses to be attenuated during feature extraction. This characteristic makes DRSN suitable for vibration-based machinery health monitoring, where fault-related components may be obscured by background vibration and measurement noise.

The principal component of DRSN is the residual shrinkage building unit (RSBU), which combines residual connections with a channel-wise thresholding mechanism. Unlike the conventional rectified linear unit (ReLU), which sets negative inputs to zero, soft-thresholding suppresses feature responses whose absolute magnitudes fall below a specified threshold and reduces the magnitudes of the remaining responses. The operation is expressed as

$$y = \operatorname{sign}(x) \max\left(|x| - \tau, 0\right)$$

where, x denotes the input feature value or feature map, y is the corresponding soft-thresholded output, and τ is a non-negative threshold parameter. The operation is applied element-wise when x represents a multidimensional feature map.

When $|x| \leq \tau$, the output becomes zero, whereas responses exceeding the threshold retain their signs with reduced magnitudes. This operation provides a mechanism for attenuating weak feature responses that may be associated with noise. Nevertheless, the relationship between feature magnitude and useful degradation information depends on the characteristics of the input signal; consequently, the threshold must be adjusted according to the learned feature representations.

A distinguishing characteristic of DRSN is that the threshold is estimated adaptively rather than assigned as a fixed parameter. In the conventional channel-wise DRSN structure, global average pooling (GAP) is applied to obtain channel descriptors, which are subsequently processed by fully connected layers to generate channel-dependent threshold coefficients. The resulting threshold vector allows different feature channels to undergo different degrees of shrinkage according to their learned responses.

For bearing RUL prediction, this mechanism provides a foundation for extracting degradation-related features from vibration signals collected under different operating conditions. However, the conventional threshold estimation structure may introduce additional parameters, and its channel interactions can be further refined. Accordingly, the DRSN-MW framework developed in this study incorporates an efficient channel attention (ECA) mechanism and multi-scale feature extraction, as described in Section 3.2. The conventional deep residual shrinkage network with channel-wise thresholding (DRSN-CW) architecture is illustrated in Figure 3.

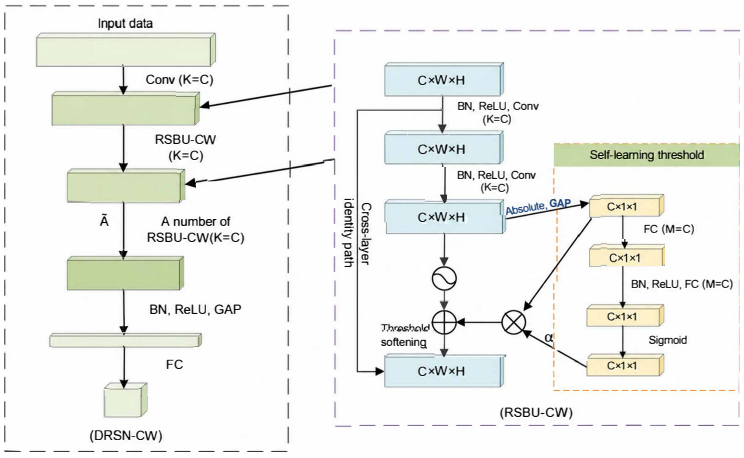


Figure 3. Architecture of the DRSN-CW

Note: DRSN-CW = deep residual shrinkage network with channel-wise thresholding; RSBU-CW = residual shrinkage building unit with channel-wise thresholding; BN = batch normalization; ReLU = rectified linear unit; GAP = global average pooling; FC = fully connected; Conv = convolution.

2.3 Fundamentals of Extension Theory

Extension theory provides a mathematical framework for representing objects, their characteristics, and the relationships between observed values and predefined characteristic ranges [31]. It is based on matter-element modeling and correlation functions that quantify the degree of association between an observation and a specified domain. In the present study, extension theory provides the theoretical basis for evaluating feature similarity between bearing datasets collected under different operating conditions.

A matter-element is represented by an ordered triplet consisting of an object, its characteristics, and the corresponding characteristic values:

$$R = (N, C, V) = \begin{bmatrix} N & c_1 & v_1 \\ & c_2 & v_2 \\ & \vdots & \vdots \\ & c_n & v_n \end{bmatrix} \tag{1}$$

where, N denotes the object under consideration, $C = [c_1, c_2, \ldots, c_n]$ is the set of n characteristics describing the object, and $V = [v_1, v_2, \ldots, v_n]$ contains their corresponding quantitative values. For cross-condition bearing analysis, these characteristics may represent degradation-related features extracted from vibration measurements, while their values describe the observed feature distributions under particular operating conditions.

To quantify the association between an observed characteristic value and a specified domain, an extension correlation function is introduced. Let $V_0 = [a, b]$ denote the classical domain, representing the characteristic interval associated with a particular condition, and let $V_p = [c, d]$ denote the joint domain, representing the broader admissible range, where $V_0 \subseteq V_p$.

The extension distance between an observed value v and the classical domain V_0 is defined as

$$\rho(v, V_0) = \left| v - \frac{a+b}{2} \right| - \frac{b-a}{2} \tag{2}$$

This distance characterizes the position of v relative to the interval boundaries. A negative value indicates that the observation lies within the classical domain, a zero value corresponds to a boundary point, and a positive value indicates that the observation lies outside the interval.

The extension correlation function is then expressed as

$$K(v) = \begin{cases} -\dfrac{\rho(v, V_0)}{|V_0|}, & v \in V_0 \\ \dfrac{\rho(v, V_0)}{\rho(v, V_p) - \rho(v, V_0)}, & v \notin V_0 \end{cases} \tag{3}$$

where, $|V_0| = b - a$ denotes the width of the classical domain. The correlation value describes the relationship between the observed characteristic and the reference interval, with larger values generally indicating a stronger association under the adopted domain definition.

In bearing degradation analysis, the extension correlation framework can be used to assess whether features extracted from different operating conditions fall within comparable characteristic ranges. However, similarity in individual feature values does not necessarily imply similarity in the temporal progression of degradation. The proposed DRSN-MW framework therefore combines extension-based feature evaluation with DTW to consider both characteristic distributions and degradation trajectories. The modified extension-based screening procedure used in the proposed framework is presented in Section 3.3.

2.4 Dynamic Time Warping

DTW is a sequence alignment method used to measure similarity between temporal observations that may differ in duration or progression rate. It is particularly relevant to machinery prognostics because bearings operating under different loads and rotational speeds may exhibit comparable degradation patterns over different time scales. DTW allows such sequences to be compared through nonlinear alignment along the temporal dimension.

Consider two feature sequences,

$$X = (x_1, x_2, \ldots, x_M)$$

and

$$Y = (y_1, y_2, \ldots, y_N)$$

where, M and N are their respective lengths. DTW determines an alignment path

$$P = (p_1, p_2, \ldots, p_K)$$

where, each path element

$$p_k = (i_k, j_k)$$

specifies a correspondence between x_{i_k} and y_{j_k}. The alignment path satisfies boundary, monotonicity, and continuity constraints, ensuring that the temporal order of observations is preserved.

The DTW distance is defined as the minimum cumulative local alignment cost over all admissible paths:

$$DTW(X, Y) = \min_{P} \sum_{k=1}^{K} d(x_{i_k}, y_{j_k}) \tag{4}$$

where, $d(\ ,\ )$ denotes the local distance function. For scalar observations, the squared Euclidean distance can be used:

$$d(a, b) = (a - b)^2$$

For multidimensional feature vectors, the corresponding squared Euclidean distance is

$$d(a, b) = \|a - b\|_2^2$$

The optimal alignment is obtained using dynamic programming. A cumulative cost matrix $D \in R^{M \times N}$ is constructed, where $D(i, j)$ represents the minimum cumulative cost of aligning the subsequences $X_{1:i}$ and $Y_{1:j}$. The recursive relationship is given by

$$D(i, j) = d(x_i, y_j) + \min \begin{cases} D(i-1, j) \\ D(i, j-1) \\ D(i-1, j-1) \end{cases} \tag{5}$$

The initial condition is defined as $D(1, 1) = d(x_1, y_1)$, while the remaining boundary conditions are assigned to prevent invalid alignment paths. The final DTW distance is obtained from $DTW(X, Y) = D(M, N)$.

By allowing nonlinear temporal correspondence, DTW can identify similarities between degradation trajectories even when their progression rates differ. Nevertheless, conventional DTW treats the selected feature dimensions according to the specified local distance function and does not inherently distinguish their relative importance to bearing degradation. In the proposed DRSN-MW framework, entropy-derived feature weights are therefore incorporated into the local distance calculation, allowing the source-domain screening procedure to account for differences in feature contributions. Combined with extension-based correlation analysis, the resulting strategy evaluates source-domain relevance from both feature-distribution and temporal-degradation perspectives, providing the basis for the multi-source screening module described in Section 3.3.

3 Proposed Method

3.1 Overview of the Proposed Framework

The proposed DRSN-MW was developed to predict the RUL of rolling bearings across different operating conditions. The framework integrates noise-robust degradation feature extraction, source-domain screening, degradation-stage-based feature alignment, and physically constrained regression within a multi-source transfer learning architecture. Labeled degradation data from multiple operating conditions are used to learn transferable representations, while unlabeled observations from the target condition are incorporated during domain adaptation. As illustrated in Figure 4, the framework comprises five interconnected modules.

(1) Shared noise-robust feature extraction. A modified DRSN was employed as the shared feature extractor $F(\ )$ to obtain degradation-related representations from raw bearing vibration signals. Adaptive soft-thresholding was combined with channel attention and multi-scale feature extraction to attenuate noise-related responses while retaining information associated with bearing deterioration. The resulting feature representations were used for source-domain screening, degradation-stage identification, and subsequent cross-domain alignment. The detailed network structure and feature extraction procedure are presented in Section 3.2.

(2) Multi-source weighted extension-based screening. A source-domain screening module was constructed to evaluate the relevance of available labeled source domains to the target operating condition. The evaluation combined weighted extension correlation, which characterizes differences in feature distributions, with entropy-weighted DTW, which measures similarities between temporal degradation trajectories. Feature weights were determined from information entropy to account for differences in the contributions of individual degradation features. Source domains with relatively low similarity to the target condition were excluded, while the selected domains were retained to form the source-domain subset $\mathcal{D}_s^{\mathrm{sel}}$. This procedure was intended to limit negative transfer caused by combining degradation data from dissimilar operating conditions. The screening method is described in Section 3.3.

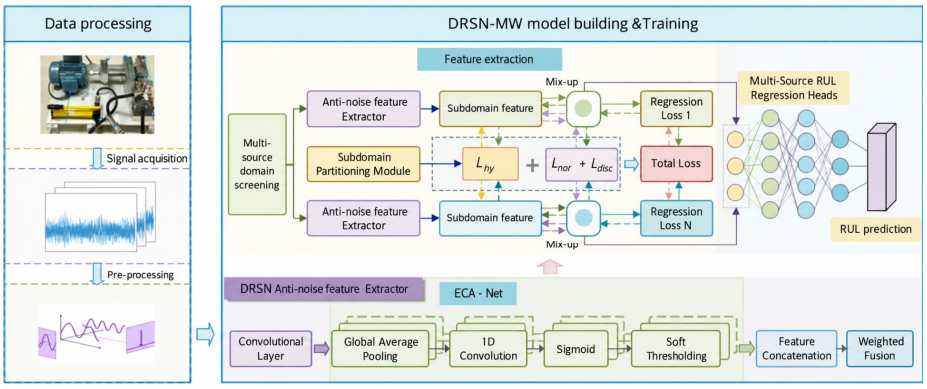


Figure 4. Architecture of the proposed DRSN-MW framework for cross-condition bearing RUL prediction

Note: DRSN-MW = deep residual shrinkage network with multi-source weighted screening; RUL = remaining useful life; DRSN = deep residual shrinkage network; ECA = efficient channel attention; DTW = dynamic time warping; LMMD = local maximum mean discrepancy.

(3) Degradation-stage partitioning. A subdomain partition network was introduced to distinguish the principal stages of bearing degradation. Following degradation onset detection, the degradation process was divided into two subdomains representing early degradation and the subsequent fault stage. The partition network was trained using labeled source-domain observations and produced stage predictions for source samples and stage probability distributions for unlabeled target samples. These outputs provided stage-related information for sample recombination and conditional feature alignment, allowing the adaptation process to account for differences in degradation progression between operating conditions.

(4) Subdomain feature alignment with sample recombination. A stage-aware alignment module was developed to reduce feature distribution discrepancies between the selected source domains and the target domain. Rather than aligning all samples within a single global feature distribution, the proposed method considered corresponding degradation stages separately. Within each subdomain, Mix-up-based feature recombination was employed to generate intermediate representations between source and target samples, thereby regularizing the regression function across domains. An entropy-weighted local maximum mean discrepancy (LMMD) loss was also introduced to measure stage-conditional distribution differences, with additional weighting assigned to samples and stages exhibiting greater uncertainty. The combination of sample recombination and conditional distribution alignment was designed to preserve degradation-stage information while reducing cross-condition feature discrepancies. The complete alignment procedure is detailed in Section 3.4.

(5) Decision optimization with consistency and physical constraints. RUL estimation was performed using multiple source-domain regression heads, denoted by $R_k(\ )$. The prediction model was trained using a supervised regression objective based on labeled source-domain samples, together with decision consistency and physical monotonicity regularization. The decision consistency term penalized discrepancies among predictions generated by different regression heads for the same target-domain observations. The monotonicity constraint penalized increases in predicted RUL between successive observations along a bearing degradation trajectory, reflecting the non-increasing nature of remaining life during continued operation. These terms were incorporated into the overall optimization objective alongside the subdomain alignment loss to regulate prediction behavior under varying operating conditions. The corresponding loss functions and optimization procedure are presented in Section 3.5.

Through these modules, DRSN-MW establishes a unified procedure for extracting degradation-related features, identifying relevant source domains, aligning stage-dependent representations, and estimating bearing RUL under previously unseen operating conditions. The framework combines information from multiple degradation datasets while accounting for differences in operating conditions and temporal degradation characteristics. Its overall architecture is illustrated in Figure 4, and the implementation of each component is described in the following sections.

3.2 Deep Residual Shrinkage Network-Based Noise-Robust Feature Extraction Module

The DRSN provides the feature extraction backbone of the proposed DRSN-MW framework. Its adaptive soft-thresholding mechanism allows low-magnitude feature responses to be attenuated during signal processing, making it suitable for extracting degradation-related information from bearing vibration signals affected by background noise and operating-condition variations [30]. However, the conventional DRSN-CW architecture estimates channel-wise thresholds primarily through global feature aggregation and fully connected transformations. Although this mechanism provides adaptive noise suppression, the estimation of channel importance can be further refined by

explicitly modeling interactions between neighboring feature channels.

To obtain more informative degradation representations for cross-condition RUL prediction, two modifications were introduced into the DRSN feature extraction module. First, an ECA mechanism was incorporated into the threshold estimation path to model local cross-channel dependencies and regulate the soft-thresholding operation. Second, a multi-scale feature extraction structure was constructed to combine representations obtained from different depths of the residual network. The resulting features were subsequently used for source-domain screening, degradation-stage partitioning, and cross-domain feature alignment.

3.2.1 Improved threshold path structure

The conventional DRSN-CW architecture employs channel-dependent soft thresholds to attenuate noise-related feature responses. These thresholds are commonly estimated using GAP followed by fully connected layers. Although effective for adaptive feature shrinkage, this approach introduces additional parameters and does not explicitly preserve local relationships between neighboring channels during threshold estimation.

To address these limitations, the efficient channel attention network (ECA-Net) [32] was incorporated into the residual shrinkage blocks. ECA-Net uses one-dimensional convolution along the channel dimension to model local cross-channel interactions without introducing a dimensionality-reduction operation. The convolutional kernel size is determined according to the number of feature channels, allowing the attention mechanism to adjust its interaction range to the network structure.

The proposed threshold adjustment procedure consisted of four steps.

(1) Channel-wise feature aggregation

Given an input feature map

$$X \in \mathbb{R}^{B\times C\times L}$$

where, B denotes the batch size, C is the number of feature channels, and L represents the temporal length of the feature sequence, GAP was applied along the temporal dimension to obtain the channel descriptor

$$z \in \mathbb{R}^{B\times C}$$

For each sample, the descriptor summarizes the average activation of each feature channel and provides the input to the subsequent attention calculation.

(2) Local cross-channel attention estimation

The channel descriptor z was passed through a one-dimensional convolutional layer to capture interactions between neighboring channels. The convolutional kernel size k was determined according to the channel dimension:

$$k = \psi(C) = \left| \frac{\log_2(C) + b}{\gamma} \right|_{\text{odd}} \tag{6}$$

where, b and γ are parameters controlling the relationship between channel dimension and kernel size, and $|\ |_{\text{odd}}$ denotes conversion to the nearest admissible odd integer.

The convolutional output was transformed using a sigmoid activation function to obtain the channel attention weights:

$$w \in (0,1)^{B\times C}$$

Each element $w_{b,c}$ represents the learned importance of channel c for sample b. Higher attention weights indicate channels receiving greater emphasis during feature processing.

(3) Adaptive threshold scaling

To regulate the degree of feature shrinkage according to channel importance, the threshold scaling coefficient was defined as

$$\alpha = 1 - w \tag{7}$$

where, w denotes the sigmoid-normalized ECA attention weights. Equivalently, if the pre-sigmoid attention output is denoted by u, the scaling coefficient can be expressed as $\alpha = 1 - \sigma(u)$.

This inverse relationship assigns smaller threshold coefficients to channels with larger attention weights, allowing more of their feature responses to be retained. Conversely, channels receiving lower attention weights are assigned larger threshold coefficients and undergo stronger shrinkage.

The mean absolute activation of each channel was then calculated as

$$m = \frac{1}{L}\sum_{l=1}^{L} |X_{:,:,l}| \tag{8}$$

where,

$$m \in \mathbb{R}^{B \times C}$$

The mean absolute activation provides a scale reference for threshold estimation, allowing the threshold magnitude to reflect the amplitude characteristics of the corresponding feature channel.

By combining the attention-derived scaling coefficient with the channel activation magnitude, the threshold estimation mechanism accounts for both channel importance and feature response intensity. This allows the degree of shrinkage to vary across channels and input samples rather than being determined by a fixed threshold.

(4) Channel-wise soft-thresholding

The adaptive threshold vector was obtained by element-wise multiplication:

$$\tau = \alpha \odot m$$

where, $\odot$ denotes element-wise multiplication.

The resulting threshold vector was expanded along the temporal dimension:

$$\tau_b = \tau[:, :, \text{None}] \in \mathbb{R}^{B \times C \times 1}$$

allowing the same threshold to be applied to all temporal positions within each channel of a given sample.

Channel-wise soft-thresholding was then performed as

$$Y = \text{sign}(X) \odot \max\left(0, |X| - \tau_b\right) \tag{9}$$

where, Y denotes the thresholded feature map, $\text{sign}(\ )$ is the element-wise sign function, and the maximum operation is applied element-wise.

Feature responses with absolute magnitudes below their corresponding thresholds were suppressed, while responses exceeding the thresholds retained their signs with reduced magnitudes. The thresholded features were subsequently combined with the residual identity mapping to obtain the output of the residual shrinkage block.

Compared with the conventional fully connected threshold estimation structure, the ECA-based mechanism introduces local cross-channel interactions through a lightweight convolutional operation. This modification was intended to reduce the parameter requirements of the threshold estimation path while retaining channel-dependent feature shrinkage. The resulting representations provide the basis for the subsequent multi-scale degradation feature extraction and cross-condition adaptation procedures.

3.2.2 Multi-scale degradation feature output

Bearing degradation develops progressively, and its vibration characteristics may change considerably throughout the operating life. During early degradation, fault-related responses can be relatively weak and difficult to distinguish from background vibration. As deterioration progresses, localized defects may produce increasingly pronounced impulsive components and periodic variations in the measured signals. At more advanced stages, changes in vibration amplitude and spectral characteristics can become more evident. These differences indicate that degradation-related information may be represented at different levels of abstraction within a deep feature extraction network.

To capture these characteristics, a four-layer DRSN architecture was constructed, and the outputs of three selected residual blocks were used to form multi-scale degradation representations. The selected feature maps were denoted by F_1, F_2, F_3, corresponding to shallow, intermediate, and deep network representations, respectively.

The shallow feature map F_1 retains relatively detailed information from the original vibration signal and can represent localized variations associated with weak fault responses. The intermediate feature map F_2 captures more abstract temporal patterns, including combinations of impulsive and periodic signal components. The deep feature map F_3 represents higher-level characteristics that may be associated with the overall progression of bearing degradation. Combining these representations allows information extracted at different network depths to be considered jointly rather than relying exclusively on the final convolutional output.

For each selected network level, the temporal dimension of the feature map was compressed using a pooling operation. GAP was applied to the shallow feature map to obtain a channel-wise representation of its average activation:

$$z_1 = \text{GAP}(F_1) \in \mathbb{R}^{C_1} \tag{10}$$

where, C_1 denotes the number of channels in the shallow feature map.

For the intermediate and deep feature maps, global max pooling (GMP) was employed to retain the largest activation in each channel. This operation was selected to preserve prominent feature responses that may correspond to localized fault-related impulses:

$$z_i = \text{GMP}(F_i) \in \mathbb{R}^{C_i}, \quad i \in \{2, 3\} \tag{11}$$

where, C_i represents the channel dimension of the corresponding feature map.

The three pooled feature vectors were concatenated to construct a unified multi-scale representation:

$$F_{\text{cat}} = \text{Concat}(z_1, z_2, z_3) \in \mathbb{R}^{C_1+C_2+C_3} \tag{12}$$

This operation preserves feature information obtained from different network depths, allowing the subsequent fusion layer to consider both relatively localized vibration responses and higher-level degradation characteristics.

To obtain a fixed-dimensional representation suitable for domain screening and feature alignment, the concatenated feature vector was passed through a fully connected transformation followed by a ReLU:

$$f(t) = \text{ReLU}(WF_{\text{cat}} + b) \in \mathbb{R}^D \tag{13}$$

where, $W \in \mathbb{R}^{D\times(C_1+C_2+C_3)}$ is the learnable weight matrix, $b \in \mathbb{R}^D$ is the bias vector, and D denotes the dimension of the fused feature representation.

The resulting feature vector combines information from the selected network levels within a common representation space. The fully connected transformation also allows the relative contributions of the pooled features to be adjusted during model training.

To preserve temporal information for subsequent DTW and degradation-stage alignment, the original vibration observations were processed using a sliding-window strategy. Each input sequence was divided into T consecutive windows, and the t-th window was transformed into a fused feature vector

$$f(t) \in \mathbb{R}^D, \quad t = 1, 2, \ldots, T$$

The resulting feature sequence was represented as

$$F = \{f(t)\}_{t=1}^{T} \in \mathbb{R}^{T\times D}$$

This sequence provides a time-ordered representation of the vibration characteristics and allows changes in degradation-related features to be examined across successive observation windows. It also provides the temporal input required for comparing degradation trajectories under different operating conditions.

To reduce variations in feature magnitude and improve numerical stability during cross-domain alignment, batch normalization was applied to the fused features, followed by L_2 normalization:

$$\widehat{f}(t) = \frac{\text{BN}(f(t))}{\|\text{BN}(f(t))\|_2} \tag{14}$$

where, $\text{BN}(\ )$ denotes batch normalization and $\|\ \|_2$ is the Euclidean norm. In numerical implementation, a small positive constant may be included in the denominator to avoid division by zero.

The normalized features were subsequently supplied to the multi-source screening and subdomain alignment modules. They also served as inputs to the regression components used for RUL estimation. Through the combination of channel-adaptive shrinkage and multi-scale feature fusion, the proposed extraction module was designed to retain degradation-related information while reducing sensitivity to noise and variations in signal amplitude across operating conditions.

3.3 Multi-Source Domain Weighted Extension-Based Screening Module

The selection of appropriate source domains is an important step in cross-condition bearing RUL prediction. Although degradation data collected under different operating conditions may contain transferable information, differences in rotational speed, mechanical load, and degradation progression can reduce the relevance of individual source domains to the target condition. Combining source data without considering these differences may introduce inconsistent degradation information and increase the risk of negative transfer.

To address this issue, a multi-source weighted extension-based screening module was developed to evaluate the relevance of available source domains before cross-domain feature alignment. The module combines feature-distribution assessment based on extension theory with temporal degradation similarity measured using entropy-weighted DTW. Information entropy is used to determine the relative contributions of individual features, while extension-based deviation analysis identifies differences between source-domain characteristics and the feature ranges observed in the target domain.

The screening procedure consisted of four stages: degradation-related feature extraction and entropy quantification, entropy-based feature weight calculation, extension correlation evaluation, and source-domain selection using entropy-weighted DTW. The selected source domains are subsequently used for subdomain alignment and RUL regression, as described in Sections 3.4 and 3.5.

3.3.1 Feature extraction and entropy quantification

The weighted extension neural network (W-ENN) [33] was developed for fault diagnosis under small-sample and noisy conditions. By incorporating feature weighting into the extension neural network, W-ENN allows differences in the contributions of individual characteristics to be considered during correlation evaluation. This principle provides a useful basis for evaluating the relative importance of degradation-related features in cross-condition bearing analysis.

In the proposed DRSN-MW framework, the feature weighting principle was adopted to quantify the information content of degradation representations extracted by the DRSN module. Rather than using the original vibration measurements directly, the screening procedure operated on the learned feature representations, which contain information extracted from different network depths. The W-ENN structure that motivated the weighting procedure is illustrated in Figure 5.

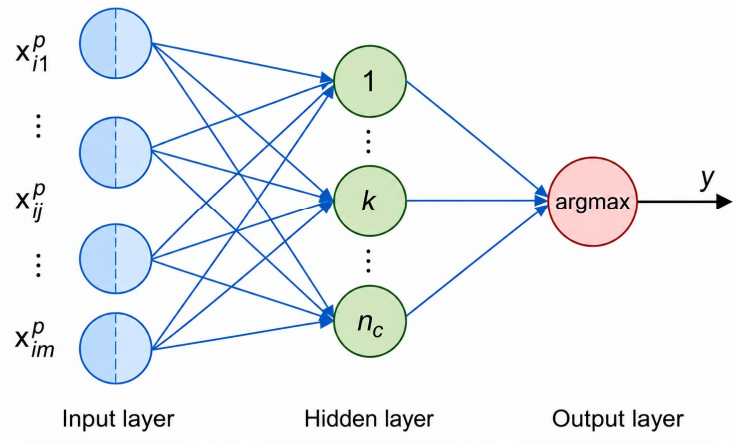


Figure 5. Architecture of the W-ENN

Note: W-ENN = weighted extension neural network.

For each extracted feature dimension, information entropy was calculated to characterize the distribution of its observed values. Let F_k denote the observations associated with the k-th feature dimension. The entropy was defined as

$$H(F_k) = -\sum_{i=1}^{N_b} p_i \log_2 p_i \tag{15}$$

where, N_b is the number of histogram bins and p_i represents the proportion of observations falling within the i-th bin. The convention $0 \log_2 0 = 0$ was adopted when a bin contained no observations.

The entropy value characterizes the distribution of observations across the selected bins. A feature concentrated within a limited number of bins generally has lower entropy, whereas a feature distributed more evenly across the bins has higher entropy. Entropy therefore provides a statistical measure of feature variability, although its relationship with degradation relevance depends on the underlying feature characteristics.

The extracted observations were organized into a feature matrix $X \in \mathbb{R}^{m \times n}$, where m denotes the number of samples and n represents the number of extracted feature dimensions. Each column of X corresponds to one feature dimension, and each row represents the feature values associated with an individual observation. This matrix provides the input for the entropy-based weighting procedure described in the following subsection.

3.3.2 Entropy-based feature weight calculation

Entropy-based weighting was employed to assign different contributions to the extracted feature dimensions during source-domain similarity evaluation. The procedure determines feature weights from the distribution of observed values without requiring manually specified importance coefficients.

Before entropy calculation, the feature values were normalized to reduce differences in numerical scale. For the j-th feature dimension, min–max normalization was performed as

$$r_{k,j} = \frac{F_{k,j} - \min(F_j)}{\max(F_j) - \min(F_j)} \tag{16}$$

where, $F_{k,j}$ denotes the value of feature j for sample k, and $\min(F_j)$ and $\max(F_j)$ are the minimum and maximum values observed for that feature in the data used to estimate its weight.

The normalized feature values were then converted into proportions:

$$p_{k,j} = \frac{r_{k,j}}{\sum_{k'=1}^{m} r_{k',j}} \tag{17}$$

where, $p_{k,j}$ represents the normalized contribution of sample k to feature dimension j. The summation index k' was used to distinguish the denominator from the sample index in the numerator.

The normalized information entropy of feature j was calculated as

$$\varepsilon_j = -\frac{1}{\ln m} \sum_{k=1}^{m} p_{k,j} \ln p_{k,j} \tag{18}$$

where, m is the number of samples used for entropy estimation. For $m > 1$, the normalization factor $\ln m$ places the entropy within the interval $[0, 1]$, provided the proportions form a valid probability distribution.

The entropy-derived weight assigned to feature j was subsequently obtained as

$$\omega_j = \frac{1 - \varepsilon_j}{\sum_{\ell=1}^{n}(1 - \varepsilon_\ell)} \tag{19}$$

where, n is the total number of feature dimensions.

The resulting weights satisfy

$$\omega_j \geq 0, \qquad \sum_{j=1}^{n} \omega_j = 1$$

Under this weighting scheme, features with lower normalized entropy receive larger weights, reflecting greater concentration or unevenness in their normalized distributions. These weights are used to control the contributions of individual feature dimensions to the subsequent source-domain similarity calculations. However, a larger entropy-derived weight should not be interpreted as direct evidence of greater physical sensitivity to bearing degradation without additional validation.

The complete feature weight vector is denoted by

$$W = [\omega_1, \omega_2, \ldots, \omega_n]$$

For numerical implementation, constant-valued features and cases involving zero normalization denominators require appropriate handling. Such features may be assigned zero weight or excluded from the weighting calculation, provided that the same procedure is applied consistently across the source and target domains.

3.3.3 Extension correlation degree evaluation

Following feature weight calculation, an extension-based similarity measure was employed to assess the relationship between each source domain and the target operating condition. The evaluation was based on the weighted deviations of source-domain features from characteristic intervals constructed using target-domain observations.

In contrast to an unweighted comparison of feature distributions, the proposed procedure incorporated the entropy-derived weights to account for differences in the contributions of individual feature dimensions. The extension-theory foundation of this approach was introduced in Section 2.3, while the weighting principle was informed by W-ENN [33].

A matter-element representation of the target domain was first constructed. Let the target-domain feature vector be expressed as

$$X_{\text{target}} = [x_{\text{target},1}, x_{\text{target},2}, \ldots, x_{\text{target},n}]$$

The corresponding target-domain matter-element model was defined as

$$M_{\text{target}} = \begin{bmatrix} O_{\text{target}} & C_1 & \langle a_{\text{target},1}, b_{\text{target},1} \rangle \\ & C_2 & \langle a_{\text{target},2}, b_{\text{target},2} \rangle \\ & \vdots & \vdots \\ & C_n & \langle a_{\text{target},n}, b_{\text{target},n} \rangle \end{bmatrix} \tag{20}$$

where, O_{target} denotes the target-domain entity, C_j represents the j-th feature characteristic, and $V_{t,j} = \langle a_{\text{target},j}, b_{\text{target},j} \rangle$ is the characteristic interval of feature j in the target domain.

The interval boundaries were determined from the corresponding target-domain feature observations:

$$a_{\text{target},j} = \min_k x_{\text{target},k,j}$$

$$b_{\text{target},j} = \max_k x_{\text{target},k,j}$$

These intervals describe the observed ranges of the target-domain features and provide reference boundaries for evaluating source-domain deviations.

To quantify deviations from the target-domain characteristic intervals, a non-negative out-of-interval distance was introduced. For a feature value x and interval $V = [a, b]$, the deviation was defined as

$$\delta(x, V) = \max\{0, a - x, x - b\} \tag{21}$$

where, $\delta(x, V) = 0$ when $x \in V$, and $\delta(x, V) > 0$ when x lies outside the interval.

This definition distinguishes observations contained within the target-domain characteristic range from those exceeding its boundaries. Unlike the conventional extension correlation function introduced in Section 2.3, the deviation measure is non-negative and directly represents the magnitude of an out-of-interval difference.

The weighted extension deviation cost for source domain k relative to the target domain was calculated as

$$\kappa_k = \sum_{j=1}^{n} \omega_j \delta(x_{k,j}, V_{t,j}) \tag{22}$$

where, $x_{k,j}$ denotes the time-averaged value of feature j in source domain k, ω_j is the corresponding entropy-derived weight, and $V_{t,j}$ is the target-domain characteristic interval for feature j.

The deviation cost was converted into a bounded similarity measure using an exponential transformation:

$$\phi_k = \exp(-\kappa_k) \tag{23}$$

where, $0 < \phi_k \leq 1$.

A smaller deviation cost corresponds to a larger similarity value. When all evaluated source-domain feature values lie within their corresponding target-domain intervals, $\kappa_k = 0$ and $\phi_k = 1$. As the weighted deviation increases, the similarity value decreases monotonically.

The resulting similarity measure provides an indication of how closely the mean feature characteristics of each source domain correspond to the observed feature ranges of the target domain. It does not, by itself, establish equivalence between complete degradation trajectories, because temporal ordering and degradation progression are not explicitly represented in the interval-based calculation.

Accordingly, extension-based similarity was used as a feature-level assessment within the source-domain screening procedure, while temporal degradation similarity was further evaluated using entropy-weighted DTW, as described in Section 3.3.4.

3.3.4 Source domain selection based on entropy-weighted dynamic time warping

Although extension-based correlation provides information about the relationship between source-domain feature values and target-domain characteristic intervals, it does not explicitly account for differences in the temporal progression of bearing degradation. Bearings operating under different rotational speeds and mechanical loads may exhibit similar degradation patterns over different time scales. Consequently, direct comparison of observations at identical time indices may not adequately reflect the similarity of their degradation trajectories.

To address this limitation, entropy-weighted DTW was incorporated into the source-domain screening procedure. DTW determines a nonlinear temporal correspondence between two feature sequences, allowing differences in degradation progression rates to be considered during similarity evaluation. The entropy-derived weights obtained in Section 3.3.2 were used to account for differences in the contributions of individual feature dimensions.

Let

$$F_s(t_i) = [F_{s,1}(t_i), \ldots, F_{s,n}(t_i)]$$

and

$$F_t(t_j) = [F_{t,1}(t_j), \ldots, F_{t,n}(t_j)]$$

denote the source-domain and target-domain feature vectors at time indices t_i and t_j, respectively.

The entropy-weighted local distance was defined as

$$d(i, j) = \sqrt{\sum_{k=1}^{n} \omega_k \left(F_{s,k}(t_i) - F_{t,k}(t_j)\right)^2} \tag{24}$$

where, ω_k is the entropy-derived weight of feature dimension k.

This weighted Euclidean distance assigns different contributions to individual feature dimensions according to their calculated weights. Features with larger weights contribute more strongly to the local alignment cost, whereas those with smaller weights have less influence on the resulting distance.

The cumulative alignment cost was calculated using dynamic programming:

$$D(i,j) = d(i,j) + \min \begin{Bmatrix} D(i-1,j) \\ D(i,j-1) \\ D(i-1,j-1) \end{Bmatrix} \tag{25}$$

where, $D(i,j)$ denotes the minimum cumulative cost of aligning the source sequence up to index i with the target sequence up to index j.

The initial condition was specified as $D(1,1) = d(1,1)$, and invalid boundary transitions were assigned infinite costs. The alignment path was constrained to preserve the temporal ordering of observations, while a warping window was used to restrict the admissible alignment region and control computational complexity.

The final entropy-weighted DTW distance between source domain k and the target domain was obtained as

$$\text{Distance}_k = D(T_s, T_t) = \text{DTW}_{\text{EW}}(\mathcal{D}_s^k, \mathcal{D}_t; W) \tag{26}$$

where, T_s and T_t denote the lengths of the source and target feature sequences, respectively, and W is the entropy-derived feature weight vector.

The admissible warping path was restricted using a predefined window parameter, which determines the maximum permitted temporal deviation between matched observations. This parameter was treated separately from the feature weight vector to avoid ambiguity in the mathematical notation.

To facilitate source-domain comparison, the DTW distance was transformed into a similarity measure:

$$\text{Similarity}_k = \frac{1}{1 + \text{Distance}_k} \tag{27}$$

where, larger values indicate smaller temporal alignment costs and therefore greater similarity between the evaluated degradation feature sequences.

A two-stage source-domain selection procedure was then applied. First, source domains satisfying $\text{Similarity}_k > \gamma$ were retained, with the screening threshold set to $\gamma = 0.7$.

The retained source domains were ranked according to their DTW-based similarity values. The M highest-ranked domains were selected, where $1 \leq M \leq N$, and the resulting source-domain subset was expressed as

$$\mathcal{D}_{\text{selected}} = \left\{ \mathcal{D}_s^{i_1} \mathcal{D}_s^{i_2} \dots \mathcal{D}_s^{i_M} \right\} \tag{28}$$

where, $i_1, i_2, \dots, i_M$ denote the indices of the selected source domains. The number of selected source domains was constrained by the number of available source domains. For single-source transfer tasks, the available source domain was retained without multi-source ranking.

The selected domains, together with the feature weight vector W, were subsequently used in the cross-domain feature alignment and RUL prediction procedures.

Overall, the proposed screening module combined entropy-based feature weighting, extension-based characteristic evaluation, and temporal degradation alignment to assess the relevance of multiple source domains to the target operating condition. Extension-based correlation characterized differences between source-domain features and target-domain reference intervals, while entropy-weighted DTW evaluated similarities in degradation progression. The resulting source-domain subset provided the training data for the stage-aware adaptation module described in Section 3.4, with the aim of reducing the influence of poorly matched operating conditions on RUL prediction.

3.4 Construction of the Subdomain Alignment Module

3.4.1 Architecture of the subdomain alignment module

Domain adaptation aims to reduce discrepancies between source and target feature distributions while preserving information relevant to the prediction task. In cross-condition bearing RUL prediction, however, the degradation process is characterized by distinct stages with different vibration responses and temporal characteristics. Conventional global alignment methods may reduce the overall distribution discrepancy without adequately preserving the correspondence between samples at similar degradation stages. As a result, observations representing different levels of deterioration may become incorrectly aligned, affecting the regression function learned for the target operating condition.

To address this limitation, a degradation-stage-aware subdomain alignment module was incorporated into the proposed DRSN-MW framework. The module divided the bearing degradation process into two subdomains, representing early degradation and the subsequent fault stage. Feature alignment was then performed within corresponding degradation stages rather than across the entire source and target distributions. This approach was

intended to retain stage-related information while reducing feature discrepancies caused by differences in operating conditions.

The two-stage partition was adopted to provide a relatively simple representation of degradation progression without introducing an excessive number of stage boundaries. The overall architecture is illustrated in Figure 6.

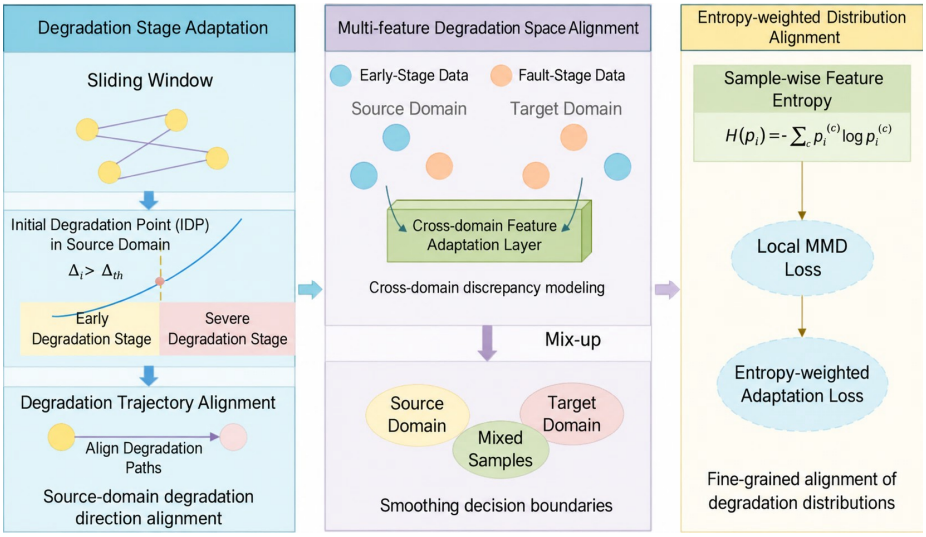


Figure 6. Workflow of the proposed degradation-stage-aware subdomain alignment module

Note: LMMD = local maximum mean discrepancy.

The module consists of three interconnected components.

Adaptive degradation-stage partitioning. An initial degradation point was identified to distinguish the degradation period from the preceding healthy operation. The remaining degradation process was subsequently divided into early degradation and fault stages. A stage classification network was trained using labeled source-domain observations, while probabilistic stage assignments were generated for unlabeled target-domain samples. These assignments provided the stage information required for conditional feature alignment.

Cross-domain feature recombination. A feature-level Mix-up strategy was employed to generate intermediate representations between source and target samples assigned to corresponding degradation stages. By restricting sample recombination to the same degradation stage, the method was designed to reduce inappropriate interpolation between observations representing substantially different health conditions. The generated features were used to regularize the regression function across operating conditions.

Entropy-weighted subdomain alignment. LMMD was employed to measure differences between source and target feature distributions within individual degradation stages. Sample-level uncertainty and stage-level entropy were incorporated into the alignment objective to adjust the contributions of different observations and degradation stages. This procedure placed greater emphasis on regions where stage assignments were less certain.

Together, these components provided a stage-aware adaptation procedure that combined degradation point detection, probabilistic stage classification, feature recombination, and conditional distribution alignment.

3.4.2 Degradation point detection and adaptive stage partitioning

Bearing vibration signals collected during the early healthy period may contain limited evidence of progressive deterioration. Including these observations without distinguishing their health states can introduce additional variation into the relationship between extracted features and RUL. Consequently, the proposed framework employed degradation point detection to identify the beginning of the deterioration period before conducting stage-wise feature alignment.

Let the normalized multi-scale feature representation obtained from the DRSN module in Section 3.2 be denoted by $h = \widehat{f}(t)$. The corresponding source-domain and target-domain representations are denoted by h^s and h^t, respectively.

(1) Degradation point detection

A sliding-window-based detection procedure was employed to identify changes in degradation-related feature trajectories. As illustrated in Figure 7, the procedure examined variations in the extracted health-related observations over successive time windows. The initial degradation point was associated with a sustained increase in the estimated degradation gradient beyond a specified threshold.

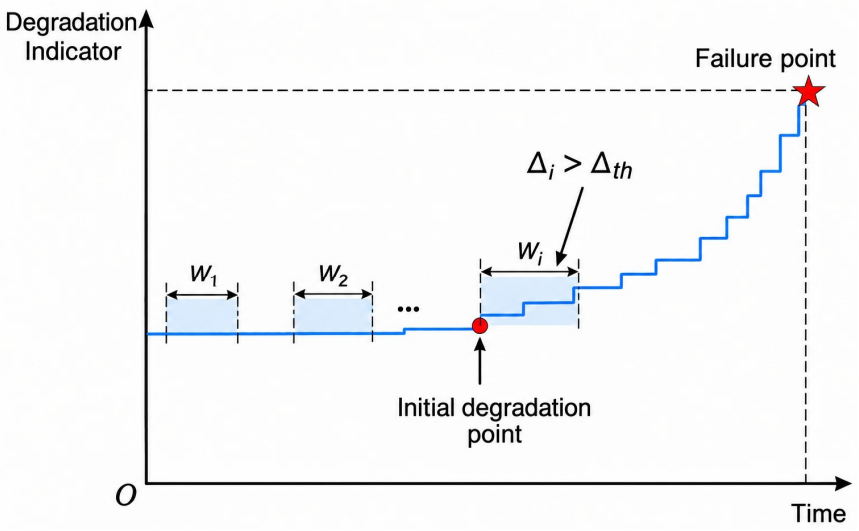


Figure 7. Schematic illustration of degradation point detection based on changes in the degradation gradient

For a sequence of degradation-related observations, a sliding window containing n consecutive points was constructed. The i-th window was defined as

$$W_i = \{y_i^*, y_{i+1}^*, \ldots, y_{i+n-1}^*\} \tag{29}$$

where, y_j^* denotes the degradation-related feature value at observation j.

The degradation gradient within each window was estimated using linear regression. The slope was calculated as

$$\Delta_i = \frac{\sum_{j \in W_i} (t_j - \bar{t}_i)(y_j - \bar{y}_i)}{\sum_{j \in W_i} (t_j - \bar{t}_i)^2} \tag{30}$$

where, t_j denotes the acquisition time and y_j represents the corresponding degradation-related observation. Here, $j \in W_i$ refers to the observation indices included in the i-th sliding window.

The mean time and mean feature value were obtained as

$$\begin{aligned} \bar{t}_i &= \frac{1}{n} \sum_{j \in W_i} t_j \\ \bar{y}_i &= \frac{1}{n} \sum_{j \in W_i} y_j \end{aligned} \tag{31}$$

The resulting gradient Δ_i characterized the average rate of change within the window. A degradation transition was identified when the estimated gradient exceeded the corresponding detection threshold.

To account for differences in signal characteristics among operating conditions, a domain-dependent threshold adjustment procedure was introduced. The initial source-domain threshold was determined from the statistical distribution of degradation gradients:

$$\Delta_{\text{th}}^{\text{source}} = \mu_{\Delta}^{\text{source}} + k\sigma_{\Delta}^{\text{source}} \tag{32}$$

where, $\mu_{\Delta}^{\text{source}}$ and $\sigma_{\Delta}^{\text{source}}$ denote the mean and standard deviation of the source-domain degradation gradients, respectively, and k is a sensitivity coefficient.

A larger value of k increases the detection threshold and therefore requires a greater change in the degradation gradient before a transition is identified.

Because the statistical characteristics of target-domain vibration signals may differ from those of the source domains, the source threshold was further adjusted using observations collected during the initial operating period of the target bearing. The target-domain threshold was calculated as

$$\Delta_{\text{th}}^{\text{target}} = \alpha \Delta_{\text{th}}^{\text{source}} + (1 - \alpha)\overline{\Delta}_{\text{target}} \tag{33}$$

where, $0 \leq \alpha \leq 1$ controls the relative contribution of the source-domain threshold, and $\overline{\Delta}_{\text{target}}$ is the mean degradation gradient estimated from the available initial target-domain observations.

This adjustment allowed the threshold to incorporate information from both the source and target operating conditions without requiring labeled target-domain RUL observations.

In addition to gradient-based detection, the Wasserstein distance was considered as a measure of distribution discrepancy between source and target features. For one-dimensional probability distributions, the first-order Wasserstein distance can be expressed as

$$W_1(P_s, P_t) = \int_{-\infty}^{+\infty} |F_s(x) - F_t(x)| \, dx \tag{34}$$

where, $F_s(x)$ and $F_t(x)$ denote the cumulative distribution functions of the source and target feature distributions, respectively.

For empirical one-dimensional distributions containing the same number of observations, the Wasserstein distance can be calculated from the ordered samples:

$$W_1(P_s, P_t) = \frac{1}{N} \sum_{i=1}^{N} \left| x_{(i)}^s - x_{(i)}^t \right| \tag{35}$$

where, $x_{(i)}^s$ and $x_{(i)}^t$ denote the i-th ordered source and target observations, respectively, and N is the number of observations in each empirical distribution.

For multidimensional feature representations, the distance can be evaluated separately along individual feature dimensions and subsequently aggregated. Such a calculation provides a dimension-wise measure of distribution discrepancy rather than the full multivariate Wasserstein distance.

The Wasserstein distance provides supplementary information about differences between source and target feature distributions. Its specific role in the degradation detection or adaptation procedure should be consistent with the implementation described in the experimental setup.

(2) Degradation-stage partitioning

After the initial degradation point was identified, the remaining degradation period was normalized to establish a common life scale across different bearings and operating conditions.

Let t_0 denote the detected degradation onset and $\mathrm{RUL}(t_0)$ represent the remaining life at that point. The normalized RUL label at time t was defined as

$$y(t) = \frac{\mathrm{RUL}(t)}{\mathrm{RUL}(t_0)} \in [0, 1] \tag{36}$$

where, $y(t_0) = 1$ corresponds to the beginning of the identified degradation period, and $y(t_f) = 0$ corresponds to the failure time t_f.

The normalized life scale was used to define two degradation stages. Observations with normalized RUL greater than 0.7 were assigned to the early degradation stage, while those with normalized RUL not exceeding 0.7 were assigned to the fault stage.

A two-layer fully connected classification network was constructed to learn the relationship between extracted degradation features and stage assignments. The network received the DRSN feature representation $h^q \in \mathbb{R}^d$ as input.

The first layer consisted of 256 fully connected units, followed by a ReLU activation function and dropout with a rate of 0.2. The second layer contained two output units followed by Softmax activation.

The stage probability vector was defined as

$$w^c = [P(c = 0 \mid h^q), P(c = 1 \mid h^q)], \quad \sum_{c=0}^{1} w^c = 1 \tag{37}$$

where, $c = 0$ denotes the early degradation stage and $c = 1$ denotes the fault stage.

The classification network was pretrained using labeled source-domain observations. The corresponding cross-entropy loss was expressed as

$$L_{\text{ce}} = -\sum_{i} \sum_{c=0}^{1} y_{i,c} \ln p_{i,c} \tag{38}$$

where, $y_{i,c}$ denotes the one-hot encoded stage label of sample i, and $p_{i,c}$ is its predicted probability of belonging to stage c.

For unlabeled target-domain observations, the trained classifier generated stage probability distributions rather than requiring known target RUL labels. These probabilities were used as soft stage assignments during subsequent feature recombination and conditional distribution alignment.

3.4.3 Subdomain-wise sample recombination

Although source and target samples may belong to corresponding degradation stages, their feature distributions can remain different because of variations in operating conditions. Direct alignment of these representations may therefore leave gaps between the source and target feature spaces.

To provide intermediate representations between corresponding source and target observations, a feature-level Mix-up strategy was incorporated into the subdomain adaptation procedure. The method generated interpolated feature vectors using samples associated with the same degradation stage.

Let h_i^s and h_j^t denote source-domain and target-domain feature vectors, respectively. For a selected pair satisfying the stage-consistency condition $c_i^s = c_j^t$, the mixed feature was calculated as

$$\widetilde{h} = \lambda h_i^s + (1-\lambda) h_j^t \tag{39}$$

where, the interpolation coefficient was sampled from a beta distribution: $\lambda \sim \mathrm{Beta}(\alpha, \alpha)$, $\alpha = 0.2$.

The interpolation coefficient controlled the relative contributions of the source and target feature vectors. Restricting interpolation to corresponding degradation stages was intended to reduce the possibility of combining representations associated with substantially different health conditions.

Because target-domain RUL labels were unavailable during training, pseudo-labels generated by the current prediction model were used when constructing regression targets for mixed samples.

The interpolated RUL label was defined as

$$\widetilde{y} = \lambda y_i^s + (1-\lambda) \widehat{y}_j^t \tag{40}$$

where, y_i^s is the known source-domain RUL label and $\widehat{y}_j^t$ denotes the predicted RUL pseudo-label for the target-domain sample.

To reduce the influence of inaccurate pseudo-labels during early training, target-domain pseudo-labels were introduced into the Mix-up procedure only during later training stages. The resulting mixed features and interpolated labels were used to regularize the regression function across source and target domains.

This procedure was intended to encourage smoother changes in predicted RUL between corresponding degradation representations while preserving the distinction between early degradation and fault stages. The resulting feature representations were subsequently incorporated into the stage-aware alignment process.

3.4.4 Entropy-weighted local maximum mean discrepancy alignment

LMMD was employed to reduce feature distribution differences between source and target domains within corresponding degradation stages. Unlike global distribution alignment, LMMD evaluates distribution discrepancies separately for each subdomain, allowing stage-related information to be retained during adaptation.

In the proposed framework, an entropy-weighted LMMD objective was constructed by incorporating sample-level uncertainty and stage-level weighting into the alignment loss. The degradation process was divided into two subdomains, and the contribution of each sample was adjusted according to the uncertainty of its predicted stage assignment.

Let the source-domain feature set be denoted by

$$\{h_i^s\}_{i=1}^{n_s}$$

and the target-domain feature set by

$$\{h_j^t\}_{j=1}^{n_t}$$

where,

$$h_i^s, h_j^t \in \mathbb{R}^D$$

The source-domain stage labels are represented by

$$c_i^s \in \{0, 1\}$$

while the target-domain stage probabilities are obtained from the partition network:

$$w_j^c = P(c \mid h_j^t)$$

To quantify the uncertainty of the predicted stage assignments, the entropy of the target-domain stage probability distribution was calculated as

$$\begin{aligned} H(w_j) &= -\sum_{c \in \{0,1\}} w_j^c \ln(w_j^c + \varepsilon) \\ u_j^t &= \frac{H(w_j)}{\ln 2} \end{aligned} \tag{41}$$

where, ε is a small positive constant introduced for numerical stability.

The normalized uncertainty weight u_j^t approaches its maximum when the two stage probabilities are similar and decreases as the classifier becomes more confident in one stage. Corresponding uncertainty weights u_i^s were obtained for source-domain samples using the stage classification network.

To account for differences in uncertainty among degradation stages, a stage-level entropy weight was introduced:

$$\eta_c = \frac{1}{|\mathcal{B}_t|} \sum_{j \in \mathcal{B}_t} w_j^c H(w_j) \tag{42}$$

where, $\mathcal{B}_t$ denotes the current target-domain training batch and $|\mathcal{B}_t|$ is its number of samples.

A larger value of η_c indicates that a greater contribution to the stage-level uncertainty measure is associated with stage c. This weight was used to adjust the relative contribution of each subdomain to the overall alignment objective.

The entropy-weighted LMMD loss was defined as

$$L_{\text{align}} = \sum_{c \in \{0,1\}} \eta_c \text{MMD}_c^2 \tag{43}$$

where, MMD_c^2 denotes the squared MMD between source and target feature distributions associated with degradation stage c.

For each subdomain, the weighted discrepancy was calculated as

$$\text{MMD}_c^2 = \sum_{i=1}^{n_s} \sum_{i'=1}^{n_s} \widetilde{a}_i^c \widetilde{a}_{i'}^c k(h_i^s, h_{i'}^s) + \sum_{j=1}^{n_t} \sum_{j'=1}^{n_t} \widetilde{b}_j^c \widetilde{b}_{j'}^c k(h_j^t, h_{j'}^t) - 2 \sum_{i=1}^{n_s} \sum_{j=1}^{n_t} \widetilde{a}_i^c \widetilde{b}_j^c k(h_i^s, h_j^t) \tag{44}$$

where, $k(\ ,\ )$ denotes the kernel function, and $\widetilde{a}_i^c$ and $\widetilde{b}_j^c$ are the normalized stage-conditional sample weights for the source and target domains, respectively.

The normalized stage-conditional sample weights were defined as

$$\begin{aligned} \widetilde{a}_i^c &= \frac{u_i^s \mathbb{I}(c_i^s = c)}{\sum_{k=1}^{n_s} u_k^s \mathbb{I}(c_k^s = c) + \varepsilon} \\ \widetilde{b}_j^c &= \frac{u_j^t w_j^c}{\sum_{k=1}^{n_t} u_k^t w_k^c + \varepsilon} \end{aligned} \tag{45}$$

where, $\mathbb{I}(\ )$ denotes the indicator function, u_i^s and u_j^t represent the uncertainty weights of source and target samples, respectively, and w_j^c is the predicted probability that target sample j belongs to degradation stage c.

The source-domain weights were determined using the known stage labels, whereas the target-domain weights incorporated probabilistic stage assignments. The normalization terms controlled the relative contributions of individual samples within each degradation stage. The small constant ε was included to avoid numerical instability when the total weight associated with a stage approached zero.

A Gaussian radial basis function kernel was adopted to evaluate nonlinear similarity between the extracted feature representations:

$$k(h, h') = \exp\left(-\gamma \|h - h'\|_2^2\right) \quad \gamma = \frac{1}{D} \tag{46}$$

where, h and h' denote feature vectors, D is the feature dimension, and γ controls the kernel bandwidth.

The kernel-based formulation allowed distribution discrepancies to be evaluated in a nonlinear feature space. By combining stage-conditional sample weights with the LMMD objective, the alignment procedure assigned different contributions to observations according to their degradation-stage membership and prediction uncertainty.

The resulting entropy-weighted LMMD loss was incorporated into the overall optimization objective of DRSN-MW. Together with stage-consistent Mix-up, the alignment module was designed to reduce distribution discrepancies between corresponding degradation stages while retaining information relevant to bearing deterioration. This provided a basis for transferring degradation representations across operating conditions without relying exclusively on global feature alignment.

3.5 Decision Optimization and Physical Constraints

Although stage-aware feature alignment reduces discrepancies between source and target representations, differences in operating conditions may still affect the regression functions learned from individual source domains. In a multi-source transfer learning framework, separate regression heads can produce different RUL estimates for the same target-domain observation. Such discrepancies may reduce prediction consistency, particularly when the available source domains exhibit different degradation characteristics.

To address this issue, the proposed DRSN-MW framework incorporated decision consistency regularization and a physical monotonicity constraint into the regression objective. The optimization procedure combined supervised RUL prediction using labeled source-domain observations with constraints on the agreement and temporal behavior of the predicted RUL values.

The supervised regression objective was constructed to minimize the discrepancy between predicted RUL values and their corresponding ground-truth labels. For N labeled source-domain observations, the regression loss was defined as

$$L_{\mathrm{reg}} = \frac{1}{N}\sum_{j=1}^{N}\left\|Y_j^s - \widehat{Y}_j^s\right\|_2^2 \tag{47}$$

where, Y_j^s denotes the ground-truth RUL label of source sample j, and $\widehat{Y}_j^s$ represents the corresponding model prediction.

This objective measures the mean squared prediction error. Minimizing L_{reg} allows the model to learn the relationship between degradation-related feature representations and RUL.

To reduce discrepancies among regression heads, a decision consistency loss was introduced. Let $F(\ )$ denote the shared feature extractor, $H_i(\ )$ the domain-specific feature transformation associated with source domain i, and $C_i(\ )$ the corresponding regression head.

For a target-domain sample x_t, the consistency loss was expressed as

$$L_{\mathrm{disc}} = \frac{2}{M(M-1)}\sum_{i<j}\left\|C_i(H_i(F(x_t))) - C_j(H_j(F(x_t)))\right\|_2 \tag{48}$$

where, M is the number of source-domain regression heads. For single-source transfer tasks with only one regression head, the decision consistency loss was set to zero. The consistency regularization was applied only when at least two source-domain regression heads were available.

The consistency loss penalizes differences between predictions generated by different regression heads for the same target-domain observation. By reducing these discrepancies, the regularization term was intended to limit inconsistencies arising from differences among source-domain degradation characteristics.

Because bearing degradation is a progressive process, the RUL is expected to decrease as operation continues toward failure. However, data-driven regression models may produce local increases in predicted RUL because of measurement noise, distribution shifts, or variations in extracted features. Such fluctuations can complicate the interpretation of predicted degradation trajectories.

To account for this characteristic, a physical monotonicity constraint was introduced. For a temporally ordered sequence of observations,

$$t_1 < t_2 < \cdots < t_B$$

the predicted RUL values were expected to satisfy

$$\widehat{y}_b \le \widehat{y}_{b-1}, \quad b = 2, \ldots, B$$

A penalty was applied whenever the predicted RUL increased between successive observations:

$$L_{\mathrm{mon}} = \frac{1}{B-1}\sum_{b=2}^{B}\mathrm{ReLU}\left(\widehat{y}_b - \widehat{y}_{b-1}\right) \tag{49}$$

where, B denotes the number of temporally ordered observations and $\mathrm{ReLU}(\ )$ is the ReLU.

When the predicted RUL decreases or remains unchanged between consecutive observations, the corresponding penalty is zero. A positive penalty is introduced only when a later observation receives a larger predicted RUL value than the preceding observation.

This regularization term was intended to discourage nonphysical increases in predicted remaining life while allowing the regression model to learn degradation patterns from the observed vibration data.

The supervised regression loss, decision consistency loss, physical monotonicity loss, and subdomain alignment loss were combined into a unified optimization objective:

$$L_{\text{total}} = L_{\text{reg}} + \lambda L_{\text{disc}} + \mu L_{\text{mon}} + \gamma L_{\text{align}} \tag{50}$$

where, the loss coefficients were set to $\lambda = 0.5$, $\mu = 1.0$, and $\gamma = 0.3$.

The coefficient λ controls the contribution of decision consistency regularization, μ determines the relative importance of the monotonicity penalty, and γ regulates the contribution of the entropy-weighted subdomain alignment loss.

The supervised regression term provided the primary prediction objective, while the additional regularization terms constrained differences among regression heads, discouraged increases in predicted RUL, and reduced stage-conditional feature distribution discrepancies.

Through the joint optimization of these objectives, the DRSN-MW framework was designed to improve the consistency of cross-condition RUL predictions while preserving degradation-related information learned from multiple source domains.

The complete training procedure, including source-domain screening, feature extraction, degradation-stage classification, subdomain alignment, and regression optimization, is summarized in Table 1.

Table 1. Training procedure of the proposed DRSN-MW framework

Step	Operation
Input	Number of training iterations T; labeled source-domain datasets $\mathcal{D}_s = \{\mathcal{D}_s^1, \ldots, \mathcal{D}_s^N\}$; unlabeled target-domain dataset $\mathcal{D}_t$.
1	Data preprocessing: Identify degradation onset points, extract degradation-stage observations, and normalize source-domain RUL labels. Calculate extension-based similarity and entropy-weighted DTW distances to obtain the selected source-domain subset $\mathcal{D}_{\text{selected}}$.
2	Stage classifier pretraining: Pretrain the subdomain partition network $Q(\,)$ using labeled source-domain observations to distinguish early degradation and fault stages.
3	Initialization: Initialize the shared feature extractor $F(\,)$, the stage partition network $Q(\,)$, and the source-domain regression heads $\{C_i(\,)\}_{i=1}^{M}$.
4	Training loop: For $t = 1, 2, \ldots, T$, perform Steps 5–11.
5	Sequence sampling: Sample temporally ordered observation sequences from $\mathcal{D}_{\text{selected}}$ and $\mathcal{D}_t$.
6	Feature extraction: Extract multi-scale representations h^s and h^t using $F(\,)$, incorporating ECA-based channel attention and adaptive soft-thresholding.
7	Stage partitioning: Generate target-domain stage probabilities using $Q(\,)$, and calculate sample-level uncertainty weights u and stage-level weights η.
8	Mix-up augmentation: Select source–target sample pairs assigned to corresponding degradation stages. Generate mixed features $\widetilde{h}$ and interpolated regression labels $\widetilde{y}$, using target-domain pseudo-labels during the designated training stage.
9	Subdomain alignment: Calculate the entropy-weighted LMMD loss L_{align} using the stage-conditional sample weights and stage-level uncertainty weights.
10	Prediction optimization: Calculate L_{reg}, L_{disc}, and L_{mon}, and obtain the total loss $L_{\text{total}} = L_{\text{reg}} + \lambda L_{\text{disc}} + \mu L_{\text{mon}} + \gamma L_{\text{align}}$.
11	Parameter update: Update the trainable parameters of the feature extraction, stage partitioning, and regression networks through backpropagation.
12	End loop: Repeat Steps 5–11 until T training iterations have been completed.
Output	Trained DRSN-MW framework for target-domain bearing RUL prediction.

Note: DRSN-MW = deep residual shrinkage network with multi-source weighted screening; RUL = remaining useful life; DTW = dynamic time warping; ECA = efficient channel attention; LMMD = local maximum mean discrepancy.

4 Experimental Results and Discussion

4.1 Experimental Setup

4.1.1 Data preprocessing

Bearing vibration signals collected under different operating conditions may exhibit substantial variations in amplitude and statistical distribution because of differences in rotational speed, mechanical load, and degradation progression. Such variations can affect feature extraction and reduce the consistency of representations learned from different operating conditions. Accordingly, a preprocessing procedure was applied before the vibration measurements were introduced into the DRSN-MW framework.

Following the degradation onset detection procedure described in Section 3.4.2, vibration observations corresponding to the identified degradation period were extracted for subsequent analysis. Min–max normalization was then applied to scale the signal amplitudes and reduce differences in numerical magnitude among the observations. The normalized vibration amplitude was calculated as

$$x_{\mathrm{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{51}$$

where, x denotes the original vibration signal amplitude, and $x_{\min}$ and $x_{\max}$ represent the minimum and maximum values used for normalization, respectively.

The normalized signals were subsequently processed using the sliding-window strategy and multi-scale feature extraction module described in Section 3.2. This procedure provided consistently scaled inputs for the source-domain screening, degradation-stage classification, and cross-domain feature alignment modules.

Min–max normalization reduces differences in numerical scale but does not, by itself, eliminate the statistical distribution discrepancies between operating conditions. These discrepancies were addressed through the multi-source screening and subdomain adaptation procedures incorporated into DRSN-MW.

4.1.2 Evaluation metrics

To evaluate RUL prediction performance, normalized remaining-life labels were constructed from the bearing degradation records. Let t denote the current time step and T the final time step corresponding to bearing failure. The normalized RUL was defined as

$$\mathrm{RUL}(t) = \frac{T - t}{T} \tag{52}$$

where, the time index is measured relative to the beginning of the evaluated degradation period. Under this definition, the normalized RUL decreases from 1 at the beginning of the degradation period to 0 at failure. The time index was measured from the detected degradation onset, so that the label definition was consistent with the normalized RUL formulation in Eq. (36).

The prediction performance was evaluated using mean absolute error (MAE) and root mean square error (RMSE). These metrics quantify the differences between predicted and reference RUL values and provide complementary measures of prediction accuracy.

Let n denote the number of test observations, $\widehat{y}_t$ the predicted RUL of observation t, and y_t its reference value. The prediction error was defined as

$$e_t = \widehat{y}_t - y_t$$

The MAE was calculated as

$$\mathrm{MAE} = \frac{1}{n}\sum_{t=1}^{n} |\widehat{y}_t - y_t| \tag{53}$$

and the RMSE was calculated as

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n} (\widehat{y}_t - y_t)^2} \tag{54}$$

Lower MAE and RMSE values indicate smaller prediction errors. MAE measures the average absolute deviation, whereas RMSE assigns greater weight to relatively large errors.

In addition to MAE and RMSE, the experimental results report a Score metric to characterize prediction performance.

All evaluation metrics were calculated using observations belonging to the identified degradation period. Healthy-stage observations preceding the detected degradation onset were excluded from the reported prediction evaluation. This restriction allowed the assessment to focus on the period during which degradation-related information was used for RUL estimation.

4.1.3 Hyperparameter settings and implementation details

The proposed DRSN-MW framework was implemented using PyTorch. Bayesian optimization was employed to determine the training hyperparameters within predefined search ranges. The batch size was searched between 16 and 64, the number of training epochs between 60 and 200, the weight decay between 0.001 and 0.1, and the initial learning rate between 0.0001 and 0.01.

The selected configuration consisted of 150 training epochs, a batch size of 64, an initial learning rate of 0.0001, and a weight decay of 0.001. The AdamW optimizer was used to update the trainable network parameters through

backpropagation. A dropout rate of 0.2 was adopted in the relevant network layers, and ReLU was used as the activation function. The sliding-window length was set to 30.

The principal training parameters are summarized in Table 2.

Table 2. Hyperparameter settings for DRSN-MW training

Parameter	Value	Parameter	Value
Training epochs	150	Weight decay	0.001
Batch size	64	Dropout rate	0.2
Initial learning rate	0.0001	Activation function	ReLU
Optimizer	AdamW	Window length	30

Note: DRSN-MW = deep residual shrinkage network with multi-source weighted screening; ReLU = rectified linear unit; AdamW = Adam with decoupled weight decay.

The same overall training configuration was used as the basis for the cross-condition transfer experiments described in the following sections. The experimental evaluation considered both single-source and multi-source transfer scenarios to examine the predictive behavior of DRSN-MW under different combinations of source and target operating conditions.

4.2 Dataset Case 1: Institute of Electrical and Electronics Engineers Prognostics and Health Management 2012 Challenge Dataset

4.2.1 Dataset description

The Institute of Electrical and Electronics Engineers (IEEE) Prognostics and Health Management (PHM) 2012 Challenge bearing dataset [34] was used to evaluate the proposed DRSN-MW framework under different operating conditions. The dataset was collected using the PRONOSTIA experimental platform, which was designed for accelerated bearing degradation testing. The platform allows bearing vibration responses to be monitored throughout the degradation process under specified rotational speeds and radial loads. The experimental arrangement is shown in Figure 8.

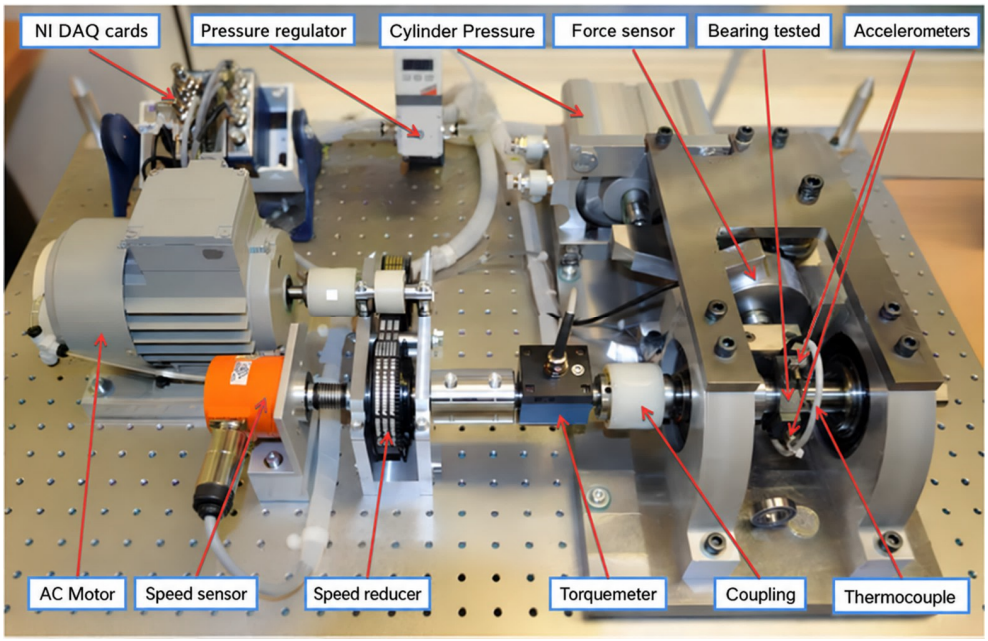


Figure 8. PRONOSTIA experimental platform used for accelerated bearing degradation testing

Note: DAQ = data acquisition; AC = alternating current.

The dataset contains vibration measurements collected from bearings operated under three conditions. Horizontal and vertical vibration signals were acquired using two DYTRAN 3035B accelerometers at a sampling frequency of 25.6 kHz, with measurements recorded at intervals of 10 s.

The operating conditions differed in rotational speed and applied load, providing a basis for investigating the influence of operating-condition changes on the transferability of degradation representations. The corresponding experimental settings are listed in Table 3.

Table 3. Operating conditions of the IEEE PHM 2012 bearing dataset

Operating Condition	Shaft Speed (rpm)	Load (N)
C1	1800	4000
C2	1650	4200
C3	1500	5000

Note: IEEE = Institute of Electrical and Electronics Engineers; PHM = prognostics and health management; C = operating condition.

The differences in rotational speed and load produce distinct vibration characteristics and degradation trajectories, making the dataset suitable for evaluating cross-condition RUL prediction methods. In the present study, the three operating conditions were treated as separate domains, and transfer learning tasks were constructed by assigning different conditions to the source and target domains.

4.2.2 Prediction task design

Six transfer learning tasks were constructed to evaluate DRSN-MW under different source-domain configurations. Tasks A–C represented single-source transfer scenarios, in which labeled degradation data from one operating condition were used to predict the RUL of bearings operating under another condition. Tasks D–F represented multi-source transfer scenarios, in which labeled data from two operating conditions were available for transfer to a third condition.

For each task, vibration observations from two bearings in each source condition were used as labeled training data. Two bearings from the target condition provided unlabeled observations for domain adaptation, while a separate target-domain bearing was reserved for testing. Target-domain RUL labels were not used during model training.

The task configurations are presented in Table 4.

Table 4. Cross-condition transfer learning task configurations for the PHM2012 dataset

Task	Source Domain	Target Domain	Labeled Training Set	Unlabeled Training Set	Test Set
A	C2	C1	B2-1, B2-2	B1-1, B1-2	B1-3
B	C3	C2	B3-1, B3-2	B2-1, B2-2	B2-3
C	C1	C3	B1-1, B1-2	B3-1, B3-2	B3-3
D	C2, C3	C1	B2-1, B2-2, B3-1, B3-2	B1-1, B1-2	B1-3
E	C1, C3	C2	B1-1, B1-2, B3-1, B3-2	B2-1, B2-2	B2-3
F	C1, C2	C3	B1-1, B1-2, B2-1, B2-2	B3-1, B3-2	B3-3

Note: PHM = prognostics and health management; C = operating condition; B = bearing.

The single-source tasks were used to examine transfer between individual operating conditions, whereas the multi-source tasks evaluated whether combining degradation information from two conditions could reduce prediction errors on the target domain.

This experimental design provided a basis for comparing source-domain configurations while maintaining a separation between labeled source-domain training data, unlabeled target-domain adaptation data, and target-domain test observations.

4.2.3 Analysis of remaining useful life prediction results on the PHM2012 dataset

The prediction performance of DRSN-MW was evaluated across the six transfer learning tasks described in Section 4.2.2. MAE, RMSE, and Score were calculated for each task, and the results are summarized in Table 5.

Table 5. RUL prediction performance of DRSN-MW on the PHM2012 dataset

Task	Source Domain	Target Domain	MAE	RMSE	Score
A	C2	C1	0.140	0.155	0.983
B	C3	C2	0.145	0.157	0.981
C	C1	C3	0.138	0.154	0.986
D	C2, C3	C1	0.112	0.147	0.990
E	C1, C3	C2	0.116	0.144	0.988
F	C1, C2	C3	0.104	0.142	0.992
Average	—	—	0.126	0.150	0.987

Note: RUL = remaining useful life; DRSN-MW = deep residual shrinkage network with multi-source weighted screening; PHM = prognostics and health management; MAE = mean absolute error; RMSE = root mean square error; C = operating condition; "—" indicates not applicable.

The results showed that DRSN-MW maintained relatively low prediction errors across the six evaluated transfer scenarios. The average MAE and RMSE were 0.126 and 0.150, respectively, while the average Score was 0.987. Among the single-source tasks, Task C produced the lowest MAE and RMSE values of 0.138 and 0.154, respectively. Task B yielded the largest prediction errors, with an MAE of 0.145 and an RMSE of 0.157.

The multi-source tasks generally produced lower prediction errors than the corresponding single-source tasks. The average MAE decreased from approximately 0.141 for Tasks A–C to 0.111 for Tasks D–F, representing a relative reduction of approximately 21.3%. The average RMSE decreased from approximately 0.155 to 0.144. These differences indicate that incorporating degradation information from multiple operating conditions was associated with improved prediction performance under the evaluated task configurations.

Task F, which transferred information from operating conditions C1 and C2 to C3, achieved the lowest MAE of 0.104 and RMSE of 0.142, together with the highest Score of 0.992. Compared with the corresponding single-source transfer task C, the MAE decreased from 0.138 to 0.104, while the RMSE decreased from 0.154 to 0.142. Similar improvements were observed when comparing Tasks A and D and Tasks B and E, which shared the same target operating conditions but differed in the number of available source domains.

These results suggest that the use of multiple source domains can provide additional degradation information for RUL prediction when the selected domains contain transferable characteristics. The weighted screening mechanism was designed to evaluate source-domain relevance, while the stage-aware alignment procedure accounted for differences in degradation progression. The individual contributions of these components are examined further through the ablation experiments presented in Section 4.2.4.

From an industrial maintenance perspective, the observed reduction in prediction errors is relevant to situations in which complete run-to-failure records are unavailable for every operating condition. The experimental results indicate the potential of transferring degradation knowledge from previously monitored bearings to equipment operating under different loads and rotational speeds. However, the present evaluation was conducted using accelerated bearing test data, and its applicability to industrial equipment operating under more variable field conditions requires further investigation.

4.2.4 Ablation study

An ablation study was conducted to examine the contributions of the principal components of the proposed DRSN-MW framework to cross-condition bearing RUL prediction. Starting from the conventional DRSN, additional components were introduced sequentially to assess changes in prediction performance. These components included degradation-stage-aware subdomain alignment, adaptive threshold adjustment, extension-based source-domain screening, and the decision optimization and physical constraints incorporated into the complete DRSN-MW framework.

The experiments were conducted using the six transfer learning tasks defined in Section 4.2.2. Prediction performance was evaluated using MAE, RMSE, and Score. The results are presented in Table 6.

The results showed a progressive reduction in prediction errors as additional components were incorporated into the baseline model. The conventional DRSN produced an average MAE of 0.343 and an average RMSE of 0.439. After introducing subdomain alignment, these values decreased to 0.275 and 0.357, respectively. The corresponding Score increased from 0.864 to 0.898. This improvement suggests that accounting for degradation-stage information during feature alignment was associated with more accurate predictions across the evaluated operating conditions.

The subsequent incorporation of adaptive threshold adjustment further reduced the average MAE from 0.275 to 0.249 and the average RMSE from 0.357 to 0.344. This result is consistent with the intended role of channel-adaptive feature shrinkage in controlling the influence of noise-related responses. The additional improvement was smaller than that observed after introducing subdomain alignment, indicating that the two modifications contributed differently within the sequential experimental configuration.

After extension-based source-domain screening was introduced, the average MAE decreased to 0.173 and the average RMSE to 0.242, while the Score increased to 0.956. The reduction in prediction errors suggests that selecting source domains according to feature similarity and degradation trajectory characteristics can improve the relevance of information transferred to the target operating condition. However, because the modules were introduced sequentially, the observed improvement reflects the combined configuration rather than an isolated measurement of the screening module.

The complete DRSN-MW framework achieved an average MAE of 0.126, an average RMSE of 0.150, and an average Score of 0.987. Relative to the baseline DRSN, the average MAE and RMSE decreased by approximately 63.3% and 65.8%, respectively. The complete model also produced the lowest prediction errors among the evaluated configurations for all six transfer tasks.

The additional reduction in errors between the source-screening configuration and the complete DRSN-MW model was consistent with the intended role of decision consistency and physical monotonicity regularization. These constraints were designed to reduce disagreements among source-domain regression heads and discourage increases in predicted RUL along the degradation trajectory. Nevertheless, the reported error metrics do not independently establish the degree to which the predicted trajectories satisfied the monotonicity constraint.

Table 6. Ablation results for the proposed DRSN-MW framework on the PHM2012 dataset

Task	Metric	DRSN	+ Subdomain Alignment	+ Threshold Adjustment	+ Source Screening	DRSN-MW
A	MAE	0.343	0.279	0.249	0.178	0.140
	RMSE	0.428	0.362	0.346	0.246	0.155
	Score	0.864	0.893	0.925	0.948	0.983
B	MAE	0.350	0.277	0.253	0.181	0.145
	RMSE	0.439	0.357	0.348	0.253	0.157
	Score	0.859	0.894	0.920	0.946	0.981
C	MAE	0.341	0.280	0.255	0.176	0.138
	RMSE	0.426	0.362	0.351	0.247	0.154
	Score	0.860	0.889	0.917	0.951	0.986
D	MAE	0.341	0.272	0.248	0.170	0.112
	RMSE	0.451	0.356	0.341	0.238	0.147
	Score	0.866	0.902	0.927	0.959	0.990
E	MAE	0.347	0.271	0.246	0.173	0.116
	RMSE	0.448	0.354	0.340	0.242	0.144
	Score	0.866	0.904	0.930	0.955	0.988
F	MAE	0.338	0.268	0.244	0.169	0.104
	RMSE	0.440	0.353	0.337	0.237	0.142
	Score	0.867	0.906	0.937	0.966	0.992
Average	MAE	0.343	0.275	0.249	0.173	0.126
	RMSE	0.439	0.357	0.344	0.242	0.150
	Score	0.864	0.898	0.926	0.956	0.987

Note: DRSN = deep residual shrinkage network; DRSN-MW = deep residual shrinkage network with multi-source weighted screening; PHM = prognostics and health management; MAE = mean absolute error; RMSE = root mean square error. The columns represent the sequential addition of model components.

The changes in MAE, RMSE, and Score across the evaluated configurations are further illustrated in Figure 9.

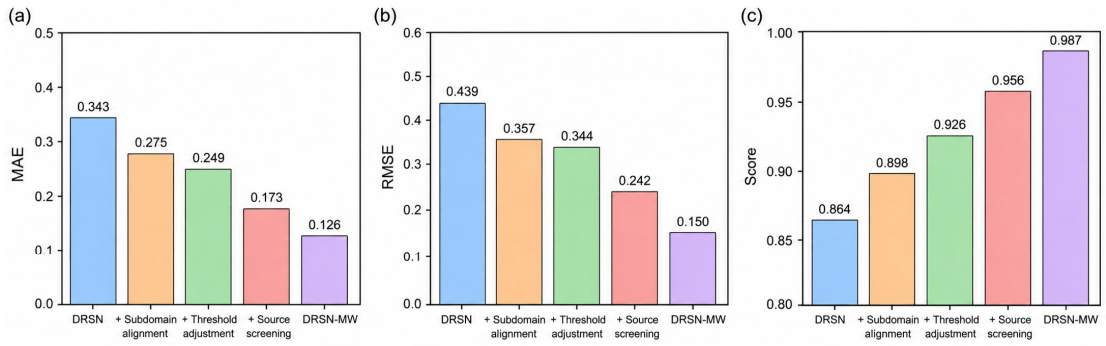


Figure 9. Prediction performance of successive DRSN-MW configurations in the ablation study: (a) MAE; (b) RMSE; (c) Score

Note: DRSN-MW = deep residual shrinkage network with multi-source weighted screening; MAE = mean absolute error; RMSE = root mean square error.

As shown in Figure 9, the sequential introduction of additional components was associated with progressively lower prediction errors and higher Score values. The improvement was observed across both single-source and multi-source transfer tasks, suggesting that the combined architecture was effective under the operating-condition combinations considered in this study.

Overall, the ablation results indicate that the complete DRSN-MW configuration provided better prediction performance than the evaluated partial configurations. The findings support the use of degradation-stage alignment, adaptive feature extraction, source-domain screening, and constrained regression within a unified cross-condition bearing prognostics framework.

4.2.5 Comparison with other methods

To assess the predictive performance of the proposed DRSN-MW framework relative to existing transfer learning approaches, comparative experiments were conducted using the six cross-condition transfer tasks defined in Section 4.2.2. The comparison included multi-prompt alignment (MPA) [35], style adaptation and uncertainty estimation (SAUE) [36], dynamic transfer (DRT) [37], and disentangling representation and adaptation network (DRANet) [38].

These methods represent different approaches to cross-domain representation learning and adaptation. MPA employs prompt-based alignment, SAUE considers adaptation and uncertainty estimation, DRT addresses multi-source domain adaptation, and DRANet separates representation learning from domain adaptation. Although these approaches were not originally developed specifically for bearing RUL prediction, they provide relevant methodological references for examining transfer learning under distribution shifts.

The original experimental design reported a common feature extraction architecture and parameter configuration for the compared methods. Such a configuration is intended to reduce differences arising from the feature extraction backbone. The reported MAE, RMSE, and Score values for the six transfer tasks are summarized in Table 7.

Table 7. Comparison of RUL prediction performance among transfer learning methods on the PHM2012 dataset

Task	Metric	MPA	SAUE	DRT	DRANet	DRSN-MW
A	MAE	0.289	0.298	0.231	0.199	0.140
	RMSE	0.351	0.362	0.284	0.253	0.155
	Score	0.875	0.891	0.931	0.921	0.983
B	MAE	0.278	0.280	0.239	0.178	0.145
	RMSE	0.331	0.337	0.293	0.225	0.157
	Score	0.889	0.914	0.928	0.943	0.981
C	MAE	0.283	0.285	0.259	0.197	0.138
	RMSE	0.342	0.347	0.318	0.252	0.154
	Score	0.881	0.906	0.919	0.927	0.986
D	MAE	0.269	0.270	0.230	0.171	0.112
	RMSE	0.321	0.326	0.283	0.218	0.147
	Score	0.903	0.923	0.934	0.954	0.990
E	MAE	0.258	0.262	0.250	0.152	0.116
	RMSE	0.312	0.316	0.305	0.190	0.144
	Score	0.909	0.930	0.931	0.961	0.988
F	MAE	0.263	0.266	0.226	0.169	0.104
	RMSE	0.318	0.321	0.277	0.215	0.142
	Score	0.907	0.929	0.925	0.951	0.992
Average	MAE	0.273	0.277	0.239	0.178	0.126
	RMSE	0.329	0.335	0.293	0.226	0.150
	Score	0.894	0.916	0.928	0.943	0.987

Note: RUL = remaining useful life; PHM = prognostics and health management; MPA = multi-prompt alignment; SAUE = style adaptation and uncertainty estimation; DRT = dynamic transfer; DRANet = disentangling representation and adaptation network; DRSN-MW = deep residual shrinkage network with multi-source weighted screening; MAE = mean absolute error; RMSE = root mean square error.

The results showed that DRSN-MW achieved lower MAE and RMSE values than the compared methods across all six transfer tasks. Its average MAE and RMSE were 0.126 and 0.150, respectively, while the average Score reached 0.987. Among the comparison methods, DRANet achieved the lowest average prediction errors, with an MAE of 0.178 and an RMSE of 0.226.

Relative to DRANet, DRSN-MW reduced the average MAE by approximately 29.2% and the average RMSE by 33.6%. The average Score increased from 0.943 to 0.987, corresponding to an absolute increase of 0.044 and a relative increase of approximately 4.7%.

The performance differences were observed in both single-source and multi-source transfer scenarios. For example, in Task A, DRSN-MW achieved an RMSE of 0.155, compared with 0.253 for DRANet. In Task F, the corresponding RMSE values were 0.142 and 0.215. These results indicate that DRSN-MW produced smaller prediction errors under the evaluated operating-condition combinations.

The observed differences may be associated with the integration of several complementary mechanisms within DRSN-MW. The source-domain screening procedure was designed to limit the transfer of poorly matched degradation information, while subdomain alignment considered differences between early and advanced degradation stages. The decision consistency and monotonicity constraints further regulated the regression outputs. The ablation results in Section 4.2.4 provide additional evidence that the complete configuration was associated with lower prediction errors than the evaluated partial configurations.

For a visual comparison, the MAE, RMSE, and Score values of the five methods across the six transfer tasks are

presented in Figure 10.

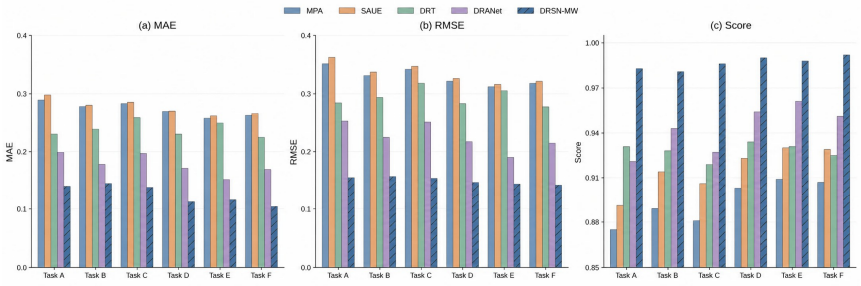


Figure 10. Comparison of RUL prediction performance among transfer learning methods across six PHM2012 tasks

Note: RUL = remaining useful life; PHM = prognostics and health management; MAE = mean absolute error; RMSE = root mean square error; MPA = multi-prompt alignment; SAUE = style adaptation and uncertainty estimation; DRT = dynamic transfer; DRANet = disentangling representation and adaptation network; DRSN-MW = deep residual shrinkage network with multi-source weighted screening.

As illustrated in Figure 10, DRSN-MW produced the lowest MAE and RMSE values in each task. Its Score values were also higher than those of the compared methods across the evaluated scenarios. The relatively small differences in prediction errors among the six tasks suggest that the proposed framework maintained consistent performance within the tested range of rotational speeds and mechanical loads.

These findings are relevant to industrial predictive maintenance because degradation data are often collected from machines operating under different conditions. A transfer learning method capable of using historical degradation information while accounting for operating-condition differences may reduce dependence on complete labeled degradation records for every target condition.

Nevertheless, the present comparison was conducted using accelerated bearing degradation datasets. The reported results therefore establish comparative performance under the specified experimental conditions rather than demonstrating field deployment performance in industrial machinery.

Overall, the comparative results indicate that DRSN-MW achieved lower prediction errors than the evaluated transfer learning methods on the PHM2012 dataset. The combination of source-domain screening, degradation-stage alignment, and constrained regression provides a plausible explanation for the observed differences, although the contribution of each mechanism should be interpreted in conjunction with the ablation results.

4.2.6 Visualization of feature distributions

To examine the changes in feature distributions before and after domain adaptation, t-distributed stochastic neighbor embedding (t-SNE) [39] was employed to visualize the high-dimensional degradation representations extracted by the shared feature extraction network. Task D, which involved transferring degradation information from operating conditions C2 and C3 to C1, was selected for visualization. The feature representations were projected into a two-dimensional space to illustrate differences in the relative distributions of source-domain and target-domain observations.

The visualization results are presented in Figure 11. Figure 11a shows the feature distributions before domain adaptation, while Figure 11b presents the corresponding distributions after applying the proposed DRSN-MW framework.

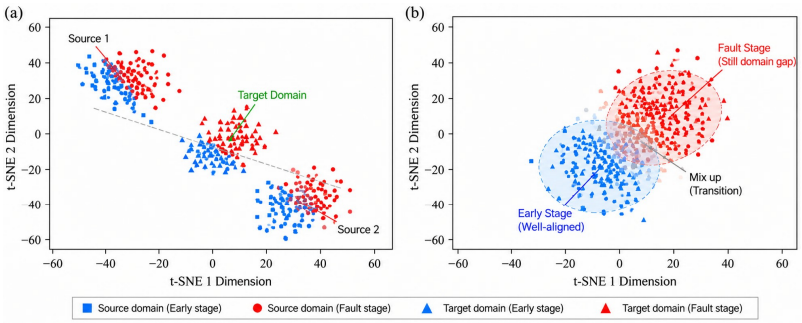


Figure 11. t-SNE visualization of bearing degradation feature distributions for Task D: (a) before domain adaptation; (b) after domain adaptation

Note: t-SNE = t-distributed stochastic neighbor embedding.

Before adaptation, the source-domain and target-domain observations occupied relatively distinct regions of the two-dimensional embedding, as shown in Figure 11a. This separation indicated differences in the extracted feature representations under the evaluated operating conditions. Such differences are consistent with the changes in vibration characteristics and degradation progression associated with variations in rotational speed and mechanical load. When source and target observations exhibit different feature distributions, transferring a regression model without adaptation may introduce errors in the estimated RUL.

After applying DRSN-MW, the projected feature distributions showed greater overlap between the source and target domains, as illustrated in Figure 11b. The separation among observations originating from different operating conditions was reduced, and the feature representations appeared more closely distributed within the embedded space. This change was consistent with the intended effect of the proposed cross-domain adaptation procedure.

Several components of DRSN-MW may have contributed to the observed distribution changes. First, the modified DRSN feature extractor employed channel-adaptive soft-thresholding and multi-scale feature fusion to obtain degradation-related representations from vibration measurements. Second, the weighted source-domain screening module evaluated the relevance of available source conditions using feature characteristics and temporal degradation similarity. Third, entropy-weighted LMMD was used to reduce distribution discrepancies between corresponding degradation stages. The stage-consistent Mix-up procedure further introduced interpolated feature representations during model training.

Together, these mechanisms were designed to reduce operating-condition-dependent variations while retaining information relevant to the progression of bearing degradation. The greater overlap observed after adaptation was consistent with this objective.

Nevertheless, the interpretation of t-SNE results requires some caution. Because t-SNE is a nonlinear dimensionality-reduction method, distances and apparent cluster overlap in the two-dimensional embedding do not directly quantify distribution discrepancies in the original high-dimensional feature space. Accordingly, the visualization provides qualitative evidence of changes in the learned representations rather than an independent quantitative measure of domain alignment.

The visualization results, considered together with the prediction errors reported in Table 5 and the ablation results in Table 6, suggest that the proposed adaptation procedure produced representations more suitable for cross-condition RUL prediction under the evaluated transfer scenario. This is relevant to industrial bearing prognostics, where vibration measurements collected under different operating conditions may exhibit substantial differences even when they describe comparable stages of equipment deterioration.

4.2.7 Noise robustness evaluation

Industrial vibration measurements are frequently affected by background interference, sensor noise, and disturbances associated with machine operation. Such interference can alter the statistical characteristics of the measured signals and affect the degradation features extracted by data-driven prediction models. Therefore, the resistance of DRSN-MW to signal noise was evaluated to examine its prediction performance under different levels of measurement interference.

White Gaussian noise was added to the normalized test signals to simulate noise contamination. The signal-to-noise ratio (SNR) was varied from -10 dB to 10 dB, covering five noise conditions:

$$\mathrm{SNR} \in \{-10, -5, 0, 5, 10\}\mathrm{dB}$$

The noise-contaminated signal was generated as

$$\widetilde{x}(t) = x(t) + \sigma n(t) \tag{55}$$

where, $x(t)$ denotes the normalized vibration signal, $n(t)$ is a zero-mean Gaussian random variable with unit variance, and σ controls the amplitude of the added noise.

The noise scaling coefficient was determined according to the specified SNR. For a signal with average power P_{signal}, the corresponding noise power is

$$P_{\mathrm{noise}} = P_{\mathrm{signal}}/10^{\mathrm{SNR}/10}$$

Because the generated Gaussian noise has unit variance, the scaling coefficient can be calculated as

$$\sigma = \sqrt{P_{\mathrm{signal}}/10^{\mathrm{SNR}/10}}$$

These auxiliary expressions clarify the relationship between the noise amplitude and the target SNR without changing the original numbering of Eq. (55).

Prediction performance was evaluated using MAE and RMSE under each noise condition. The results were compared with those obtained using the original signals without added noise, as summarized in Table 8.

Table 8. RUL prediction errors under different SNRs

Metric	−10 dB	−5 dB	0 dB	5 dB	10 dB	Original
MAE	0.240	0.197	0.165	0.147	0.134	0.126
RMSE	0.288	0.221	0.190	0.169	0.153	0.150

Note: RUL = remaining useful life; SNR = signal-to-noise ratio; MAE = mean absolute error; RMSE = root mean square error.

The results showed that prediction errors increased as the SNR decreased. Under the original noise-free evaluation condition, DRSN-MW achieved an MAE of 0.126 and an RMSE of 0.150. At an SNR of 10 dB, the MAE increased to 0.134 and the RMSE to 0.153, corresponding to relative increases of approximately 6.3% and 2.0%, respectively.

When the SNR decreased to 5 dB, the MAE and RMSE increased to 0.147 and 0.169. Relative to the original evaluation condition, these values represented increases of approximately 16.7% and 12.7%, respectively. Although prediction accuracy declined, the increase in error remained smaller than that observed under the more severe noise conditions.

At an SNR of 0 dB, the MAE reached 0.165 and the RMSE increased to 0.190. Further reductions in SNR produced larger prediction errors. At -5 dB, the MAE and RMSE were 0.197 and 0.221, respectively, while the most severe noise condition of -10 dB resulted in an MAE of 0.240 and an RMSE of 0.288.

These results indicate that the proposed model retained a degree of resistance to moderate Gaussian noise contamination, although its prediction accuracy deteriorated substantially under stronger interference. The gradual increase in prediction errors suggests that the feature extraction and adaptation procedures continued to provide useful degradation information over the evaluated SNR range, but their ability to distinguish degradation-related responses became more limited as the noise intensity increased.

The observed behavior is consistent with the design of the DRSN-based feature extraction module. Channel-adaptive soft-thresholding was introduced to attenuate low-magnitude feature responses, while multi-scale feature fusion combined information from different network depths. The source-domain screening and subdomain alignment components were designed to reduce differences among feature representations associated with varying operating conditions. However, the present experiments do not independently establish the contribution of each component to noise resistance.

From an industrial predictive maintenance perspective, the results are relevant because bearing vibration signals are rarely collected under completely noise-free conditions. The evaluation suggests that DRSN-MW can maintain comparatively low prediction errors under moderate simulated interference, while stronger noise contamination remains a limitation.

It should also be recognized that the present robustness evaluation was based on additive white Gaussian noise. Actual industrial vibration measurements may contain impulsive interference, nonstationary disturbances, electromagnetic noise, and other forms of contamination that differ from the simulated conditions. The reported results therefore characterize robustness to the specified Gaussian noise levels rather than to all types of industrial measurement interference.

Overall, the noise robustness experiments showed a systematic relationship between signal quality and RUL prediction error. The findings provide additional evidence of the behavior of DRSN-MW under measurement disturbances and complement the cross-condition prediction results presented in the preceding sections.

4.3 Case 2: Xi'an Jiaotong University–Sumyoung Bearing Dataset

4.3.1 Dataset description and task configuration

To further evaluate the cross-condition RUL prediction performance of the proposed DRSN-MW framework, experiments were conducted using the XJTU-SY bearing dataset. This publicly available dataset contains vibration measurements collected throughout accelerated bearing degradation tests under different rotational speeds and mechanical loads. The dataset provides an additional experimental basis for examining the transferability of degradation representations across operating conditions.

The vibration measurements were acquired using an accelerated bearing life test platform consisting primarily of an alternating current (AC) motor, support bearings, a rotational speed controller, and acceleration sensors. The experimental arrangement is illustrated in Figure 12.

Horizontal and vertical acceleration sensors were mounted on the bearing housing to record vibration responses during operation. LDK-UER204 bearings were used in the experiments, and their principal geometric and mechanical parameters are summarized in Table 9.

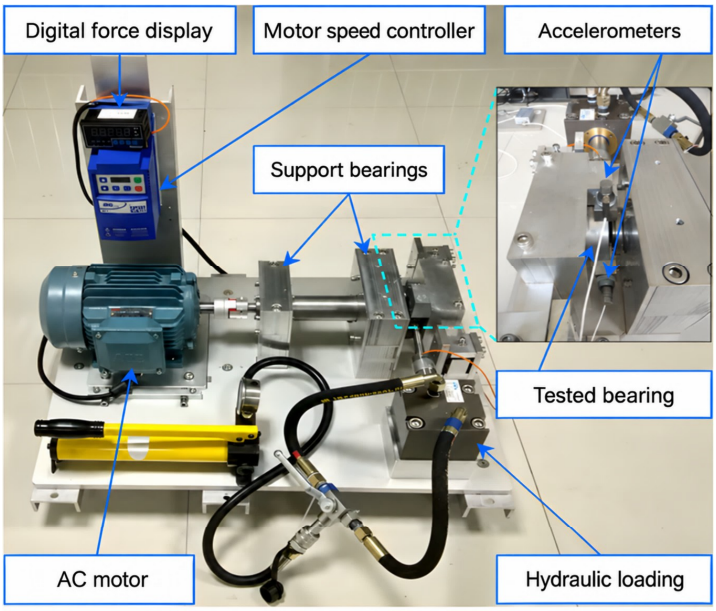


Figure 12. Experimental platform used for accelerated bearing life testing in the XJTU-SY dataset

Note: XJTU-SY = Xi'an Jiaotong University–Sumyoung; AC = alternating current.

Table 9. Geometric and mechanical parameters of the bearings in the XJTU-SY dataset

Parameter	Value	Parameter	Value
Outer race diameter (mm)	39.80	Inner race diameter (mm)	29.30
Pitch diameter (mm)	34.55	Ball diameter (mm)	7.92
Number of balls	8	Contact angle (°)	0
Static load (kN)	6.65	Dynamic load (kN)	12.82

Note: XJTU-SY = Xi'an Jiaotong University–Sumyoung.

The vibration signals were acquired at a sampling frequency of 25.6 kHz, with a recording duration of 1.28 s for each acquisition and a sampling interval of 1 min. The accelerated tests continued until the specified termination criterion was reached. According to the dataset description used in this study, data acquisition was terminated when the measured acceleration amplitude exceeded 20 g, indicating an advanced fault condition.

The XJTU-SY dataset contains complete degradation records from 15 bearings tested under three operating conditions, with five bearings assigned to each condition. The operating conditions differed in rotational speed and applied load, providing distinct experimental settings for investigating cross-condition degradation prediction. The corresponding operating parameters are listed in Table 10.

Table 10. Operating conditions of the XJTU-SY bearing dataset

Parameter	C1	C2	C3
Load (kN)	12	11	10
Rotational speed (rpm)	2,100	2,250	2,400
Number of bearings	5	5	5

Note: XJTU-SY = Xi'an Jiaotong University-Sumyoung; C = operating condition.

The differences in rotational speed and mechanical load provided a basis for evaluating whether degradation information learned from one or more operating conditions could be transferred to another condition. In particular, the dataset allowed the proposed framework to be assessed using a different experimental platform from that employed in the PHM2012 evaluation.

Six cross-condition transfer tasks were constructed, comprising three single-source tasks and three multi-source tasks. For the single-source tasks, labeled observations from one operating condition were used to train the prediction model, while unlabeled observations from the target condition were incorporated into the domain adaptation procedure. For the multi-source tasks, labeled degradation observations from two operating conditions were available for source-domain screening and transfer learning.

In each task, the labeled source-domain training data and unlabeled target-domain adaptation data were separated from the target-domain test bearing. The target-domain test observations were reserved for performance evaluation, and their RUL labels were not used for supervised model training.

The detailed task configurations are presented in Table 11.

Table 11. Cross-condition transfer learning task configurations for the XJTU-SY dataset

Task	Source Domain	Target Domain	Labeled Training Set	Unlabeled Training Set	Test Set
A	C1	C2	B1-1, B1-2	B2-1, B2-2	B2-3
B	C2	C3	B2-1, B2-2	B3-1, B3-2	B3-3
C	C3	C1	B3-1, B3-2	B1-1, B1-2	B1-3
D	C1, C2	C3	B1-1, B1-2, B2-1, B2-2	B3-1, B3-2	B3-4
E	C2, C3	C1	B2-1, B2-2, B3-1, B3-2	B1-1, B1-2	B1-4
F	C1, C3	C2	B1-1, B1-2, B3-1, B3-2	B2-1, B2-2	B2-4

Note: XJTU-SY = Xi'an Jiaotong University-Sumyoung; C = operating condition; B = bearing.

The task configurations allowed the performance of DRSN-MW to be examined under different source–target combinations. Tasks A–C assessed transfer from individual source conditions, whereas Tasks D–F examined the use of multiple source domains. The latter configuration was particularly relevant to the proposed source-domain screening mechanism, which was designed to identify degradation information more closely related to the target operating condition.

The XJTU-SY experiments therefore provided a complementary evaluation of DRSN-MW using an independent bearing degradation dataset, allowing the consistency of prediction performance across the two experimental platforms to be examined.

4.3.2 Analysis of remaining useful life prediction results on the Xi'an Jiaotong University–Sumyoung dataset

The proposed DRSN-MW framework was evaluated using the six cross-condition transfer tasks defined in Section 4.3.1. Each task was repeated independently three times, and the average prediction results were reported using MAE, RMSE, and Score. The experimental results are summarized in Table 12.

Table 12. RUL prediction performance of DRSN-MW on the XJTU-SY dataset

Task	Source Domain	Target Domain	MAE	RMSE	Score
A	C1	C2	0.135	0.162	0.983
B	C2	C3	0.133	0.159	0.984
C	C3	C1	0.138	0.167	0.982
D	C1, C2	C3	0.122	0.149	0.989
E	C2, C3	C1	0.126	0.152	0.987
F	C1, C3	C2	0.119	0.145	0.991
Average	—	—	0.129	0.156	0.986

Note: RUL = remaining useful life; DRSN-MW = deep residual shrinkage network with multi-source weighted screening; XJTU-SY = Xi'an Jiaotong University-Sumyoung; MAE = mean absolute error; RMSE = root mean square error; C = operating condition; "—" indicates not applicable.

The results showed that DRSN-MW maintained relatively low prediction errors across the six evaluated operating-condition combinations. The average MAE and RMSE were 0.129 and 0.156, respectively, while the average Score was 0.986. These results were close to those obtained using the PHM2012 dataset, for which the corresponding average MAE, RMSE, and Score were 0.126, 0.150, and 0.987.

Among the single-source transfer tasks, Task A produced the lowest MAE of 0.135, whereas Task B achieved the lowest RMSE of 0.159. Task C yielded the largest prediction errors, with an MAE of 0.138 and an RMSE of 0.167. These differences indicate that prediction performance varied according to the source–target operating-condition combination, even when the same transfer learning framework was applied.

The multi-source tasks produced lower prediction errors than the single-source tasks in the reported configurations. For Tasks A–C, the average MAE and RMSE were approximately 0.135 and 0.163, respectively. For Tasks D–F, the corresponding values decreased to approximately 0.122 and 0.149. These differences represented relative reductions of approximately 9.6% in MAE and 8.6% in RMSE.

Task F, which transferred degradation information from conditions C1 and C3 to C2, achieved the lowest prediction errors, with an MAE of 0.119 and an RMSE of 0.145. It also obtained the highest Score of 0.991. Task D, which used conditions C1 and C2 as source domains and C3 as the target domain, achieved an MAE of 0.122 and an RMSE of 0.149. Task E produced similarly low prediction errors, with an MAE of 0.126 and an RMSE of 0.152.

The reduction in prediction errors observed in the multi-source tasks suggests that incorporating degradation information from multiple operating conditions can improve RUL estimation when the available source data contain transferable characteristics. In the proposed DRSN-MW framework, the weighted screening procedure was designed to assess the relevance of source-domain features and degradation trajectories, while the subdomain alignment module considered differences between corresponding degradation stages.

These mechanisms may have contributed to the observed performance differences. However, the present comparison involved different combinations of source domains and target bearings, and the reported results should therefore be interpreted as evidence of performance under the evaluated configurations rather than as an isolated measurement of the contribution of source-domain screening.

A comparison of the results obtained from PHM2012 and XJTU-SY showed relatively small differences in the average evaluation metrics. The average MAE differed by 0.003, the average RMSE by 0.006, and the average Score by 0.001. This consistency indicates that DRSN-MW maintained similar levels of normalized prediction error across the two bearing datasets.

Nevertheless, the datasets differed in experimental platform, operating conditions, and bearing degradation characteristics. Similar average errors do not necessarily imply identical generalization behavior, particularly because the evaluation metrics were calculated using normalized RUL values. The results instead suggest that the proposed framework was applicable to the two evaluated accelerated bearing degradation datasets.

From an industrial predictive maintenance perspective, the XJTU-SY experiments provide additional evidence that degradation knowledge obtained under previously monitored operating conditions can be used to support RUL prediction under different loads and rotational speeds. This capability is relevant to industrial rotating machinery, where complete run-to-failure records are often unavailable for every operating condition.

Overall, the XJTU-SY results showed that DRSN-MW achieved consistent prediction performance across the evaluated transfer tasks. The findings complement the PHM2012 experiments and support further investigation of multi-source transfer learning for industrial bearing health prognostics.

4.3.3 Comparison with other methods

To further examine the predictive performance of DRSN-MW on the XJTU-SY dataset, comparative experiments were reported using conventional deep learning models and transfer learning approaches. The comparison included adaptability feature perception fusion model (AFPFM), cross-residual attention transformer (CRAformer), MPA, SAUE, DRT, and DRANet.

These methods represented different approaches to feature learning and cross-domain adaptation. AFPFM and CRAformer were included as deep learning comparison models, while MPA, SAUE, DRT, and DRANet provided transfer learning references. The latter approaches were originally developed for broader domain adaptation problems and were considered here in the context of bearing RUL prediction.

The original experimental description identified Task D as the representative transfer scenario selected for comparison. The reported results are presented in Table 13.

Table 13. Comparative RUL prediction results on the XJTU-SY dataset

Model Category	Model	MAE	RMSE	Score
Deep learning	AFPFM	0.347	0.404	0.854
Deep learning	CRAformer	0.324	0.387	0.876
Transfer learning	MPA	0.282	0.323	0.896
Transfer learning	SAUE	0.264	0.317	0.913
Transfer learning	DRT	0.240	0.286	0.932
Transfer learning	DRANet	0.183	0.231	0.959
Proposed framework	DRSN-MW	0.133	0.159	0.984

Note: RUL = remaining useful life; XJTU-SY = Xi'an Jiaotong University–Sumyoung; MPA = multi-prompt alignment; SAUE = style adaptation and uncertainty estimation; DRT = dynamic transfer; DRANet = disentangling representation and adaptation network; DRSN-MW = deep residual shrinkage network with multi-source weighted screening; MAE = mean absolute error; RMSE = root mean square error; AFPFM = adaptability feature perception fusion model; CRAformer = cross-residual attention transformer.

The reported results showed that DRSN-MW achieved the lowest MAE and RMSE among the evaluated methods. Its MAE was 0.133 and its RMSE was 0.159, while the corresponding Score reached 0.984.

Among the conventional deep learning models, AFPFM produced an MAE of 0.347 and an RMSE of 0.404, whereas CRAformer achieved an MAE of 0.324 and an RMSE of 0.387. The transfer learning approaches generally produced lower prediction errors than these two models under the reported comparison conditions.

Among the transfer learning methods, DRANet achieved the lowest MAE and RMSE, with values of 0.183 and 0.231, respectively. Compared with DRANet, DRSN-MW reduced MAE by approximately 27.3% and RMSE by

31.2%. The Score increased from 0.959 to 0.984, corresponding to an absolute increase of 0.025.

The differences between the methods are consistent with the importance of accounting for distribution discrepancies when transferring degradation information across operating conditions. Conventional feature learning methods may capture useful vibration characteristics but do not necessarily account for differences between source and target distributions. Domain adaptation approaches address this problem by incorporating mechanisms intended to reduce such discrepancies.

The proposed DRSN-MW framework combined source-domain screening, degradation-stage-aware feature alignment, and constrained regression within a unified prediction architecture. The screening procedure evaluated source-domain relevance, while the entropy-weighted LMMD module considered distribution differences between corresponding degradation stages. Decision consistency and physical monotonicity regularization were further introduced to control the behavior of the regression outputs.

These components provide possible explanations for the lower prediction errors observed in the reported comparison. Nevertheless, the contribution of each mechanism cannot be established from Table 13 alone and should be interpreted together with the ablation results presented in Section 4.2.4.

When considered alongside the PHM2012 results, the XJTU-SY comparison suggests that DRSN-MW can produce lower normalized RUL prediction errors than the evaluated approaches under the reported experimental conditions. The findings support the relevance of multi-source transfer learning and degradation-stage alignment to bearing health prognostics, while further evaluation under variable industrial operating conditions remains necessary.

5 Conclusions

This study developed a multi-source weighted transfer learning framework, termed DRSN-MW, for RUL prediction of rolling bearings under different operating conditions. The framework integrated a DRSN with adaptive threshold adjustment, extension-based source-domain screening, degradation-stage-aware subdomain alignment, and physically constrained regression. Entropy-weighted DTW was employed to assess the similarity of degradation trajectories across operating conditions, while Mix-up-based feature recombination and entropy-weighted LMMD were introduced to reduce distribution discrepancies between corresponding degradation stages. Decision consistency regularization and physical monotonicity constraints were further incorporated to regulate the predicted RUL trajectories. These components were designed to address the combined effects of heterogeneous degradation patterns, source-domain differences, and prediction inconsistency in cross-condition bearing prognostics.

The proposed framework was evaluated using the PHM2012 and XJTU-SY bearing datasets under single-source and multi-source transfer scenarios. On the PHM2012 dataset, DRSN-MW achieved average MAE and RMSE values of 0.126 and 0.150, respectively, with an average Score of 0.987. Compared with DRANet, the strongest comparison method in terms of average prediction error, the proposed framework reduced MAE by 29.2% and RMSE by 33.6%. On the XJTU-SY dataset, average MAE and RMSE values of 0.129 and 0.156 were obtained, respectively, with an average Score of 0.986. The multi-source transfer tasks generally produced lower prediction errors than the single-source tasks on both datasets. The ablation results further showed that the complete DRSN-MW configuration achieved lower prediction errors than the evaluated partial configurations. In addition, the noise robustness experiments indicated that prediction accuracy was relatively stable under moderate Gaussian noise contamination, although larger errors were observed at lower SNRs.

The results indicate that combining source-domain screening, degradation-stage alignment, and physical constraints is a useful approach to cross-condition bearing RUL prediction. By transferring degradation information from previously monitored operating conditions, the proposed framework provides a basis for predictive maintenance of industrial rotating machinery when labeled run-to-failure data are limited for the target condition. Nevertheless, the present evaluation was based on accelerated bearing degradation datasets, and the effects of continuously varying operating conditions, different machinery configurations, and complex industrial measurement noise remain to be investigated. Further validation using operational monitoring data is therefore required to establish the applicability of DRSN-MW to industrial maintenance systems.

Author Contributions

Conceptualization, Y.L.C. and L.Q.Z.; methodology, Y.L.C. and L.Q.Z.; software, Y.L.C.; validation, Y.L.C. and L.Q.Z.; formal analysis, Y.L.C.; investigation, Y.L.C.; resources, L.Q.Z.; data curation, Y.L.C.; writing—original draft preparation, Y.L.C.; writing—review and editing, Y.L.C. and L.Q.Z.; visualization, Y.L.C.; supervision, L.Q.Z.; project administration, Y.L.C. and L.Q.Z.; funding acquisition, L.Q.Z. All authors have read and agreed to the published version of the manuscript.

Funding

This work is funded by the National Key Laboratory of Helicopter Aeromechanics Foundation, China (Grant No.: 2023-HA-LB-067-07) and the Jiangsu Province Natural Science Foundation, China (Grant No.: BK20221481).

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

References

[1] Y. Lei, N. Li, L. Guo, N. Li, T. Yan, and J. Lin, "Machinery health prognostics: A systematic review from data acquisition to RUL prediction," *Mech. Syst. Signal Process.*, vol. 104, pp. 799–834, 2018. https://doi.org/10.1016/j.ymssp.2017.11.016

[2] G. Yu and Z. Cui, "A high-resolution time-frequency analysis tool for fault diagnosis of rotating machinery," *Chin. J. Mech. Eng.*, vol. 39, p. 100014, 2026. https://doi.org/10.1016/j.cjme.2025.100014

[3] B. Wang, Y. Lei, N. Li, and N. Li, "A hybrid prognostics approach for estimating remaining useful life of rolling element bearings," *IEEE Trans. Reliab.*, vol. 69, no. 1, pp. 401–412, 2020. https://doi.org/10.1109/TR.2018.2882682

[4] S. Fu and N. P. Avdelidis, "Prognostic and health management of critical aircraft systems and components: An overview," *Sensors*, vol. 23, no. 19, p. 8124, 2023. https://doi.org/10.3390/s23198124

[5] X. Chen, R. Yang, Y. Xue, M. Huang, R. Ferrero, and Z. Wang, "Deep transfer learning for bearing fault diagnosis: A systematic review since 2016," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–21, 2023. https://doi.org/10.1109/TIM.2023.3244237

[6] H. Gu, R. Wang, K. Zhang, J. Wang, L. Sun, K. Li, X. Zhang, and S. Tu, "Fatigue reliability modeling using an energy-based parameter by loading cycle-failure life interference: Validation and application," *Chin. J. Mech. Eng.*, vol. 39, p. 100082, 2026. https://doi.org/10.1016/j.cjme.2025.100082

[7] Y. Du, X. Geng, B. Zhang, Q. Zhou, S. Cheng, and H. Zhang, "Novel bearing fault diagnosis model for wind turbine: PSR-CNN-DLSTM combined with transfer learning," *Chin. J. Mech. Eng.*, vol. 39, p. 100204, 2026. https://doi.org/10.1016/j.cjme.2025.100204

[8] S. Du, C. Zhang, S. Wang, R. Chen, Y. Zhang, and R. Mu, "Remaining useful life prediction for hydraulic spool valve based on physics-informed Gamma process," *Chin. J. Aeronaut.*, vol. 39, no. 3, p. 103824, 2026. https://doi.org/10.1016/j.cja.2025.103824

[9] J. Chen, R. Huang, Z. Chen, W. Mao, and W. Li, "Transfer learning algorithms for bearing remaining useful life prediction: A comprehensive review from an industrial application perspective," *Mech. Syst. Signal Process.*, vol. 193, p. 110239, 2023. https://doi.org/10.1016/j.ymssp.2023.110239

[10] X. Dong, C. Zhang, H. Liu, D. Wang, and T. Wang, "A multi-constrained domain adaptation network for remaining useful life prediction of bearings," *Mech. Syst. Signal Process.*, vol. 206, p. 110900, 2024. https://doi.org/10.1016/j.ymssp.2023.110900

[11] W. Zhang, X. Li, and Q. Ding, "Deep residual learning-based fault diagnosis method for rotating machinery," *ISA Trans.*, vol. 95, pp. 295–305, 2019. https://doi.org/10.1016/j.isatra.2018.12.025

[12] B. Wang, Y. Lei, N. Li, and T. Yan, "Deep separable convolutional network for remaining useful life prediction of machinery," *Mech. Syst. Signal Process.*, vol. 134, p. 106330, 2019. https://doi.org/10.1016/j.ymssp.2019.106330

[13] J. Zhu, N. Chen, and C. Shen, "A new deep transfer learning method for bearing fault diagnosis under different working conditions," *IEEE Sens. J.*, vol. 20, no. 15, pp. 8394–8402, 2020. https://doi.org/10.1109/JSEN.2019.2936932

[14] X. Li, W. Zhang, Q. Ding, and X. Li, "Diagnosing rotating machines with weakly supervised data using deep transfer learning," *IEEE Trans. Ind. Inform.*, vol. 16, no. 3, pp. 1688–1697, 2020. https://doi.org/10.1109/TII.2019.2927590

[15] Y. Zhu, F. Zhuang, J. Wang, G. Ke, J. Chen, J. Bian, H. Xiong, and Q. He, "Deep subdomain adaptation network for image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 4, pp. 1713–1722, 2021. https://doi.org/10.1109/TNNLS.2020.2988928

[16] J. Li, R. Huang, G. He, S. Wang, G. Li, and W. Li, "A deep adversarial transfer learning network for machinery emerging fault detection," *IEEE Sens. J.*, vol. 20, no. 15, pp. 8413–8422, 2020. https://doi.org/10.1109/JSEN.2020.2975286

[17] T. Han, C. Liu, W. Yang, and D. Jiang, "Deep transfer network with joint distribution adaptation: A new intelligent fault diagnosis framework for industry application," *ISA Trans.*, vol. 97, pp. 269–281, 2020. https://doi.org/10.1016/j.isatra.2019.08.012

[18] Z. Chen, M. Wu, R. Zhao, F. Guretno, R. Yan, and X. Li, "Machine remaining useful life prediction via an attention-based deep learning approach," *IEEE Trans. Ind. Electron.*, vol. 68, no. 3, pp. 2521–2531, 2021. https://doi.org/10.1109/TIE.2020.2972443

[19] X. Li, W. Zhang, Q. Ding, and J.-Q. Sun, "Intelligent rotating machinery fault diagnosis based on deep learning using data augmentation," *J. Intell. Manuf.*, vol. 31, no. 2, pp. 433–452, 2020. https://doi.org/10.1007/s10845-018-1456-1

[20] T. Hu, Y. Guo, L. Gu, Y. Zhou, Z. Zhang, and Z. Zhou, "Remaining useful life prediction of bearings under different working conditions using a deep feature disentanglement based transfer learning method," *Reliab. Eng. Syst. Saf.*, vol. 219, p. 108265, 2022. https://doi.org/10.1016/j.ress.2021.108265

[21] M. Huang and C. Zhang, "A novel multi-source domain adaptation method with Dempster–Shafer evidence theory for cross-domain classification," *Mathematics*, vol. 10, no. 15, p. 2797, 2022. https://doi.org/10.3390/math10152797

[22] Y. Liu, B. Hou, M. Ahmed, Z. Mao, J. Feng, and Z. Chen, "A hybrid deep learning approach for remaining useful life prediction of lithium-ion batteries based on discharging fragments," *Appl. Energy*, vol. 358, p. 122555, 2024. https://doi.org/10.1016/j.apenergy.2023.122555

[23] X. Zheng, Z. He, J. Nie, P. Li, Z. Dong, and M. Gao, "A progressive multi-source domain adaptation method for bearing fault diagnosis," *Appl. Acoust.*, vol. 216, p. 109797, 2024. https://doi.org/10.1016/j.apacoust.2023.109797

[24] J. Liu, C. Pan, F. Lei, D. Hu, and H. Zuo, "Fault prediction of bearings based on LSTM and statistical process analysis," *Reliab. Eng. Syst. Saf.*, vol. 214, p. 107646, 2021. https://doi.org/10.1016/j.ress.2021.107646

[25] A. Y. Zhou and A. Barati Farimani, "FaultFormer: Pretraining transformers for adaptable bearing fault classification," *IEEE Access*, vol. 12, pp. 70 719–70 728, 2024. https://doi.org/10.1109/ACCESS.2024.3399670

[26] M. A. Munir, M. H. Khan, M. S. Sarfraz, and M. Ali, "Domain adaptive object detection via balancing between self-training and adversarial learning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 12, pp. 14 353–14 365, 2023. https://doi.org/10.1109/TPAMI.2023.3290135

[27] T. Chen, S.-H. Wang, Q. Wang, Z. Zhang, G.-S. Xie, and Z. Tang, "Enhanced feature alignment for unsupervised domain adaptation of semantic segmentation," *IEEE Trans. Multimedia*, vol. 24, pp. 1042–1054, 2022. https://doi.org/10.1109/TMM.2021.3106095

[28] D. Peng, H. Guan, Y. Zang, and L. Bruzzone, "Full-level domain adaptation for building extraction in very-high-resolution optical remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022. https://doi.org/10.1109/TGRS.2021.3093004

[29] Y. Deng, S. Du, D. Wang, Y. Shao, and D. Huang, "A calibration-based hybrid transfer learning framework for RUL prediction of rolling bearing across different machines," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–15, 2023. https://doi.org/10.1109/TIM.2023.3260283

[30] M. Zhao, S. Zhong, X. Fu, B. Tang, and M. Pecht, "Deep residual shrinkage networks for fault diagnosis," *IEEE Trans. Ind. Inform.*, vol. 16, no. 7, pp. 4681–4690, 2020. https://doi.org/10.1109/TII.2019.2943898

[31] W. Cai and C. Yang, "Basic theory and methodology on Extenics," *Chin. Sci. Bull.*, vol. 58, no. 13, pp. 1190–1199, 2013. https://doi.org/10.1360/972012-1472

[32] Q. Wang, B. Wu, P. Zhu, P. Li, W. Zuo, and Q. Hu, "ECA-Net: Efficient channel attention for deep convolutional neural networks," in *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Seattle, WA, USA, 2020, pp. 11 531–11 539. https://doi.org/10.1109/CVPR42600.2020.01155

[33] T. Wang, J. Wang, Y. Wu, and X. Sheng, "A fault diagnosis model based on weighted extension neural network for turbo-generator sets on small samples with noise," *Chin. J. Aeronaut.*, vol. 33, no. 10, pp. 2757–2769, 2020. https://doi.org/10.1016/j.cja.2020.06.024

[34] P. Nectoux, R. Gouriveau, K. Medjaher, E. Ramasso, B. Chebel-Morello, N. Zerhouni, and C. Varnier, "PRONOSTIA: An experimental platform for bearings accelerated degradation tests," in *2012 IEEE International Conference on Prognostics and Health Management*, Denver, Colorado, United States, 2012, pp. 1–8. https://hal.science/hal-00719503

[35] H. Chen, X. Han, Z. Wu, and Y. G. Jiang, "Multi-prompt alignment for multi-source unsupervised domain adaptation," in *Advances in Neural Information Processing Systems*, New Orleans, Louisiana, USA, 2023, pp. 74 127–74 139. https://doi.org/10.52202/075280-3243

[36] Y. Lu, H. Huang, and X. Hu, "Style adaptation and uncertainty estimation for multi-source blended-target domain adaptation," in *Advances in Neural Information Processing Systems*, Vancouver, Canada, 2024, pp. 87 042–87 060. https://doi.org/10.52202/079017-2762

[37] Y. Li, L. Yuan, Y. Chen, P. Wang, and N. Vasconcelos, "Dynamic transfer for multi-source domain adaptation," in *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Nashville, TN, USA, 2021, pp. 10 993–11 002. https://doi.org/10.1109/CVPR46437.2021.01085

[38] S. Lee, S. Cho, and S. Im, "DRANet: Disentangling representation and adaptation networks for unsupervised cross-domain adaptation," in *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Nashville, TN, USA, 2021, pp. 15 247–15 256. https://doi.org/10.1109/CVPR46437.2021.01500

[39] L. van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 86, pp. 2579–2605, 2008. https://www.jmlr.org/papers/v9/vandermaaten08a.html